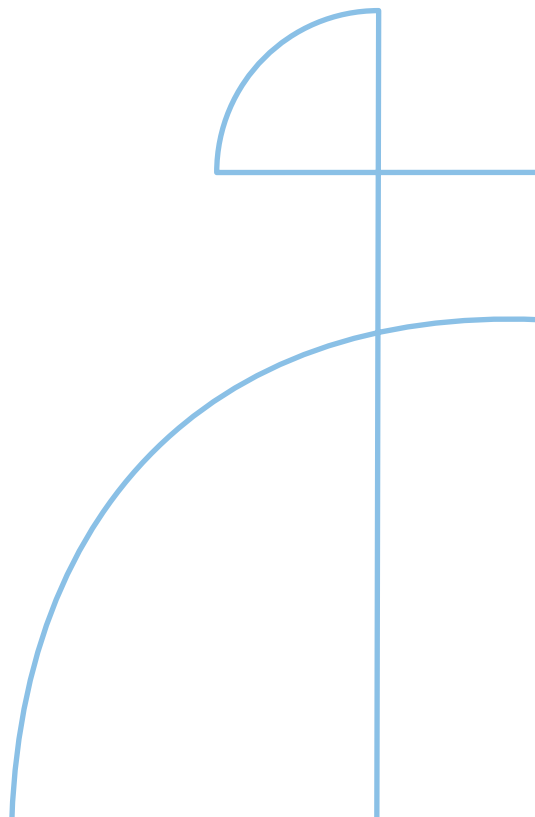


Doctoral Thesis in Information and Communication Technology

Analytical Foundations of Energy-Efficient Wireless Communication: RIS Control and Sleep Mode Management

ANDERS ENQVIST

KTH ROYAL INSTITUTE OF TECHNOLOGY



Analytical Foundations of Energy-Efficient Wireless Communication: RIS Control and Sleep Mode Management

ANDERS ENQVIST

Academic Dissertation which, with due permission of the KTH Royal Institute of Technology, is submitted for public defence for the Degree of Doctor of Philosophy on Monday, the 7th of September 2026, at 13:00, F3, Kungl. Tekniska högskolan, Lindstedtsvägen 26, Stockholm.

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Abstract

This thesis explores the optimization of energy efficiency (EE) and practical configuration overhead in wireless communication systems, focusing on both the user equipment (UE) and the base station (BS).

The first part of the study examines minimizing the UE's energy consumption when transmitting short data payloads via a BS-controlled reconfigurable intelligent surface (RIS). To balance the energy cost of the additional pilot signals needed to configure the RIS against the energy savings during data transmission, we propose dividing the RIS into controllable subarrays. This yields a unique energy-minimizing configuration determined by payload size and path loss conditions. Building on the practical challenges of RIS implementation, we then address the wireless control signaling overhead required to feed back these phase-shift configurations to the RIS. We propose a novel quantization codebook that guarantees minimal signal-to-noise ratio (SNR) loss while reducing feedback overhead from linear to logarithmic scaling with the number of RIS elements, and we introduce an efficient differential feedback scheme for mobility scenarios.

The second part focuses on the EE of multi-antenna BSs using an active and passive transceiver model. By jointly optimizing transmit power, bandwidth, and the number of antennas, we derive novel closed-form solutions that uncover a fundamental relationship between radiated power and passive circuitry consumption. This optimization framework is extended to incorporate Quality-of-Service constraints and advanced sleep modes, leading to a dynamic scheduling algorithm that optimizes hardware configurations to minimize total energy consumption for bursty traffic.

Finally, the thesis investigates the fundamental EE limits of a dual-band BS site combining a coverage-oriented sub-6 GHz carrier with a high-bandwidth,

capacity-oriented secondary carrier. By jointly optimizing hardware parameters and sleep-mode activity across both bands, we identify four distinct operational regions that govern EE-optimal behavior. We derive analytical thresholds dictating precisely when the secondary capacity band should awaken from sleep and how much traffic it should carry, demonstrating that its utilization becomes EE-optimal only when the power cost of the bandwidth-limited primary band exceeds the static power penalty of activating the secondary circuitry.

Together, these studies provide a comprehensive view of EE optimization and resource management, offering novel theoretical insights and actionable design guidelines for UE and BS configurations, RIS control signaling, sleep-mode management, and multi-band integration.

Keywords
6G, Energy Efficiency, Reconfigurable Intelligent Surfaces, Advanced Sleep Modes, Control Channel

Sammanfattning

Denna avhandling utforskar optimeringen av energieffektivitet (EE) och den signaleringsoverhead som krävs i trådlösa kommunikationssystem, med fokus på både användaren (UE) och basstationen (BS).

Den första delen av studien undersöker hur energiförbrukningen hos en UE kan minimeras vid överföring av små datamängder via en rekonfigurerbar intelligent yta (RIS) styrd av BS. För att balansera energikostnaden för de extra pilotsignaler som krävs för att konfigurera ytan mot energibesparingarna under dataöverföringen föreslår vi att RIS:ens element delas in i styrbara grupper (subarrays). Detta ger en unik energiminimerande konfiguration som bestäms av datamängdens storlek och radiokanalens dämpning. Med utgångspunkt i de praktiska utmaningarna vid RIS-implementering behandlar vi därefter den trådlösa kontrollsignalering som krävs för att återkoppla dessa faskonfigurationer till RIS:en. Vi föreslår en ny kvantiseringskodbok som garanterar en minimal förlust i signal-brusförhållande (SNR) samtidigt som signaleringsoverheaden reduceras från att skala linjärt till att skala logaritmiskt med antalet RIS-element. Dessutom introducerar vi ett effektivt differentiellt kontrollsignaleringsschema för mobilitetsscenarier.

Den andra delen fokuserar på EE hos basstationer med flera antenner, med hjälp av en effektförbrukningsmodell som tar hänsyn till sändarens aktiva och passiva komponenter. Genom att gemensamt optimera sändareffekt, bandbredd och antal antenner härleder vi nya analytiska lösningar som visar ett samband mellan utstrålad effekt och passiv kretsförbrukning. Detta optimeringsramverk utvidgas vidare till att omfatta tjänstekvalitetskrav (QoS) och avancerade vilolägen, vilket leder till en dynamisk schemaläggningsalgoritm som anpassar hårdvarukonfigurationen för att minimera den totala energiförbrukningen vid skurvis trafik.

Slutligen undersöker avhandlingen de grundläggande gränserna för energieffektivitet hos en tvåbandsbasstation som kombinerar en täckningsorienterad primär bärvåg under 6 GHz med en kapacitetsorienterad sekundär bärvåg med hög bandbredd. Genom att gemensamt optimera hårdvaruparametrar och vilolägenas aktivitet över de båda banden identifierar vi fyra driftsområden för EE överföring. Vi härleder analytiska tröskelvärden som exakt anger när det sekundära kapacitetsbandet bör väckas ur sitt viloläge, och visar att aktiveringen är optimal ur energieffektivitetssynpunkt först när kostnaden att tvinga ytterligare trafik genom det bandbreddsbegränsade primära bandet överstiger den statiska effektkostnaden för att aktivera det sekundära bandet.

Sammantaget ger dessa studier en samlad bild av energieffektiv optimering och resurshantering. De bidrar med nya teoretiska insikter och praktiska riktlinjer för utformningen av BS-konfigurationer, RIS-kontrollsignalering, hantering av vilolägen och multibandsintegration.

Nyckelord
6G, energieffektivitet, rekonfigurerbara intelligenta ytor, avancerade vilolägen, kontrollkanal

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I extend my profound appreciation to my main supervisor, Professor Emil Björnson, whose guidance, expertise, and support have been the cornerstone of my research. His knowledge in the field and his ability to articulate complex concepts in an understandable manner have been invaluable. His mentorship has also significantly contributed to my growth as a researcher.

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Lastly, I would like to express my gratitude to my family and friends for their love and encouragement. Their belief in me has been a source of strength throughout this journey.

Anders Enqvist
Stockholm, August 2026

List of included papers

Contributions This doctoral dissertation consists of two parts. The first part provides an overview of the research field on which I focused and a summary of my contributions to it. The second part comprises the four appended papers through which I have made these contributions. All four papers are co-authored with Özlem Tuğfe Demir, Cicek Cavdar, and Emil Björnson.

- **Paper A: Optimizing Reconfigurable Intelligent Surfaces for Short Transmissions: How Detailed Configurations can be Afforded?**
  Published in: *IEEE Transactions on Wireless Communications*, vol. 23, no. 4, pp. 3377–3391, 2023.
- **Paper B: Minimal Feedback Control Signaling for Reconfigurable Intelligent Surfaces: Codebook Design and SNR Analysis.**
  Published in: *IEEE Transactions on Wireless Communications*, vol. 25, pp. 15134–15148, 2026.
- **Paper C: Fundamentals of Energy-Efficient Hardware Configurations for Wireless Links with Sleep Modes.**
  Submitted to: *IEEE Transactions on Wireless Communications*, 2026.
- **Paper D: Energy-Efficient Dual-Band Communication: How to Allocate Traffic to Sub-THz Carriers?**
  Presented at: *The 7th International Conference on Communications, Signal Processing, and their Applications (ICCSPA'26)*, Spain, 2026.

Other contributions The following conference papers by the author are related to the appended journal papers but are not included in the thesis. They are the

conference versions of Papers A, B, and C, respectively.

- **Optimizing Reconfigurable Intelligent Surfaces for Small Data Packets: A Subarray Approach.**
  Published in: *IEEE International Conference on Communications (ICC)*, Seoul, South Korea, pp. 2664–2669, 2022.
- **Control Signaling for Reconfigurable Intelligent Surfaces: How Many Bits are Needed?**
  Published in: *IEEE International Conference on Communications (ICC)*, Montreal, Canada, pp. 6856–6861, 2025.
- **Fundamentals of Energy-Efficient Wireless Links: Optimal Ratios and Scaling Behaviors.**
  Published in: *IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, Singapore, pp. 1–6, 2024.

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List of Acronyms

3GPP	Third Generation Partnership Program
4G	Fourth generation
5G	Fifth generation
5QI	5G quality-of-service identifier
6G	Sixth generation
ADC	Analog-to-digital converter
BB	Baseband
BPF	Band pass filter
BS	Base station
BWP	Bandwidth part
CA	Carrier aggregation
CDF	Cumulative distribution function
CRS	Cell-specific reference signal
CSI	Channel state information
DAC	Digital-to-analog converter
dB	Decibel
DBS	Dual-band saturation
DFT	Discrete Fourier transform
DL	Downlink
DSP	Digital signal processor
EC	Energy consumption
EE	Energy efficiency
EM	Electromagnetic
FDD	Frequency division duplex
GHz	Gigahertz
HPBW	Half-power beamwidth
IF	Intermediate frequency
i.i.d.	Independent and identically distributed
IoT	Internet of things
IPP	Interrupted Poisson process
ITU	International Telecommunication Union
KKT	Karush-Kuhn-Tucker
LO	Local oscillator
LoS	Line of sight

LPF	Low pass filter
LTE	Long Term Evolution (4G)
MEMS	Micro-electro-mechanical-system
MIMO	Multiple-input multiple-output
MISO	Multiple-input single-output
mmW	Millimeter wave
MN	Mobile network
MNO	Mobile network operator
MRC	Maximum ratio combining
MRT	Maximum ratio transmission
NLoS	Non line of sight
NOMA	Non-orthogonal multiple access
NR	New Radio (5G)
OFDM	Orthogonal frequency division multiplexing
PA	Power amplifier
PBDC	Primary-band duty-cycling
PBS	Primary-band saturation
PC	Power consumption
PDB	Packet delay budget
PHY	Physical layer
PIN	Positive-intrinsic-negative (diode)
QAM	Quadrature amplitude modulation
QoS	Quality of service
RAN	Radio access network
RF	Radio frequency
RIS	Reconfigurable intelligent surface
SAR	Specific absorption rate
SBDC	Secondary-band duty-cycling
SISO	Single-input single-output
SNR	Signal-to-noise ratio
SRS	Sounding reference signal
SSB	Synchronization signal block
THz	Terahertz
UE	User equipment
UL	Uplink
ULA	Uniform linear array
UPA	Uniform planar array
URLLC	Ultra-reliable low-latency communication
WPT	Wireless power transfer

Notation

In this thesis, sets are denoted by upper-case script letters, e.g., $\mathcal{S}$. The cardinality of a set $\mathcal{S}$ is represented by $|\mathcal{S}|$. Vectors are denoted by lower-case bold-face letters, e.g., $\mathbf{x}$, and matrices by upper-case bold-face letters, e.g., $\mathbf{X}$. We denote random variables with upper-case letters, e.g., X. Also, we use $f_X(x)$ to indicate the probability density function (PDF) of a continuous random variable X at x.

$\mathbb{E}[\cdot]$	Expectation operator.
$\mathbb{P}\{\cdot\}$	Probability operator.
$\mathcal{N}(\mu, \sigma^2)$	Gaussian distribution with mean μ and variance σ^2.
$\mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$	Circular symmetric complex Gaussian distribution with mean μ and variance σ^2.
$\mathbb{C}$	Set of complex numbers.
$\lvert \cdot \rvert$	Absolute value of a scalar, or cardinality of a set.
$\lVert \cdot \rVert$	Euclidean norm.
$\triangleq$	Equal by definition.
$(\cdot)^{\mathrm{T}}$	Transpose.
$(\cdot)^{\mathrm{H}}$	Conjugate transpose.
$\mathrm{diag}(\cdot)$	Diagonal matrix formed from a vector.
$W_0(\cdot)$	Principal branch of the Lambert W function.
$W_{-1}(\cdot)$	Lower branch of the Lambert W function.
$\lceil \cdot \rceil$	Ceiling operator.
$\arg\max$	Maximizing argument.
$(\cdot)^\star$	Optimal value of a variable.
$a \ll b$	a is much less than b.
$a \gg b$	a is much greater than b.
$a \in \mathcal{A}$	a is a member of $\mathcal{A}$.
$\mathrm{Re}\{z\}$	Real part of a complex number z.

Introduction

1 Introduction

The data usage in wireless networks grows at a rapid rate which has resulted in a significant increase in energy consumption (EC) [8]. If this increased EC is to be kept in check, wireless transmission technologies must also become more energy efficient (EE). With the ever-increasing demand for higher data rates, understanding how to make wireless communication systems consume less energy has become crucial [9]. In the rapidly evolving field of wireless communications, EE has emerged as a paramount concern, ensuring that technological advancement and sustainable practices go hand in hand. The introduction of massive MIMO (multiple-input multiple-output) [10] has greatly improved data rates in communication systems, and the introduction of new technologies such as reconfigurable intelligent surfaces (RIS) [11] has the potential to further reduce EC in the future. In recent times, EE has also received increased attention with targets now being defined by the ITU [12]. Operators have also expressed great interest in this area [13].

The individual wireless links might not be rate-constrained in the sixth generation (6G) wireless technology, thanks to enormous bandwidths, but rather limited by EC. By understanding the fundamental limitations of EE in wireless communication systems, we can identify new approaches and technologies that can lead to more sustainable and efficient wireless networks in the future.

In LTE (Long Term Evolution - 4G) networks, the paper [14] revealed that the idle power consumption (PC) was very high because the energy was consumed regardless of whether the network is used to transmit data to a UE or not. This implies that the PC is mostly independent of the traffic load and highlights the vast potential for energy savings by improving the EE of BSs, particularly in scenarios with little load on the network.

This thesis investigates the possible gains in EE as measured in bits transmitted per unit of energy (Joule) in wireless communication links utilizing multiple antenna technologies. It studies the effects of utilizing RIS to increase the EE in realistic scenarios with finite payloads, where the objective is to optimize the EE under various constraints such as having a certain maximum allowed transmit power, bandwidth or number of transmit antennas. To achieve this, we implement realistic models of the environment, the hardware, and also the relevant design parameters in wireless BSs. This includes but is not limited to MIMO setup, pilot configuration and modeling hardware components, and the impacts of imperfect channel state information (CSI) and phase shift quantization in the RIS. Sequential optimization of certain performance indicators is employed to develop converging algorithms for design choices. How the transmit power for both pilot and data signaling may be adapted to the load variations in the cell when utilizing an RIS and its impact on the overall EE is also studied.

1.1 Papers Included in the Thesis

Four research papers on topics related to EE are included in this thesis.

- **Paper A: Optimizing Reconfigurable Intelligent Surfaces for Short Transmissions: How Detailed Configurations can be Afforded?**
  Published in: *IEEE Transactions on Wireless Communications*, vol. 23, no. 4, pp. 3377–3391, 2023.
- **Paper B: Minimal Feedback Control Signaling for Reconfigurable Intelligent Surfaces: Codebook Design and SNR Analysis.**
  Published in: *IEEE Transactions on Wireless Communications*, vol. 25, pp. 15134–15148, 2026.
- **Paper C: Fundamentals of Energy-Efficient Hardware Configurations for Wireless Links with Sleep Modes.**
  Submitted to: *IEEE Transactions on Wireless Communications*, 2026.
- **Paper D: Energy-Efficient Dual-Band Communication: How to Allocate Traffic to Sub-THz Carriers?**
  Presented at: *The 7th International Conference on Communications, Signal Processing, and their Applications (ICCSPA'26)*, Spain, 2026.

Paper A studies how to minimize the energy consumption of a UE that transmits a short data payload via a BS-controlled RIS, accounting for the pilot overhead needed to configure it. To balance the energy cost of the pilot signals against the energy savings during data transmission, we propose grouping the RIS elements into subarrays that share a common phase shift, and we jointly optimize the subarray size and the pilot power as a function of the payload length and

the path losses on the direct and RIS-assisted links. We show that this problem admits a unique energy-minimizing solution, and we characterize how the optimal configuration changes with the operating regime. Short payloads and strong direct paths favor large subarrays with low pilot power, while long payloads and weaker direct paths favor finer per-element control with higher pilot power.

Paper B addresses the wireless control signaling overhead required to feed back RIS phase-shift configurations from the BS to the RIS. We design a novel quantization codebook that exploits the geometric structure of the cascaded LoS channel, reducing the feedback load from linear to logarithmic scaling with the number of RIS elements while guaranteeing that the resulting beam remains within the half-power beamwidth. We extend the codebook design to three-dimensional planar RIS geometries and to channels with a non-negligible static path, and propose a low-overhead differential feedback scheme of only a few bits per update for mobility scenarios. We further analyze the combined SNR loss arising from finite feedback resolution and discrete phase-shift hardware.

Paper C combines state-of-the-art hardware models with a mathematical analysis of the upper limitations of EE. Therein we discover new relationships that hold between transmit power, number of transmit antennas, bandwidth, and the hardware characterizing constants. We also analytically derive upper limits on the resulting EE as well as develop an algorithm for optimizing the parameters of this specific non-linear optimization problem. We then extend the framework to incorporate Quality-of-Service (QoS) requirements and advanced sleep modes, characterizing how latency constraints and sleep-mode transition times shape the EE-optimal hardware configuration. Building on this, we propose a dynamic event-driven scheduling algorithm that manages a packet buffer with absolute deadlines, switching between the optimal active configuration and different sleep states to minimize total energy consumption under bursty stochastic traffic.

Paper D investigates the EE limits of a dual-band BS site that combines a coverage-oriented sub-6 GHz primary carrier with a high-bandwidth, capacity-oriented sub-THz secondary carrier. By jointly optimizing the transmit power, number of active antennas, and sleep-mode activity factors across both bands, we identify four distinct operational regions that govern the EE-optimal behavior of the system. We derive analytical thresholds, expressed in closed form via the Lambert W function, that dictate precisely when the secondary band should be awakened from sleep, and show that its activation is EE-optimal only once the power cost of the bandwidth-limited primary band exceeds the static power penalty of the secondary circuitry. We further establish the asymptotic optimal traffic-allocation rule between the two bands in the very high-rate regime.

The mapping between the appended papers and the thesis chapters is as fol-

lows. Paper A is summarized in Chapter 3, Paper B in Chapter 4, and Paper D in Chapter 8. The contributions of Paper C span two chapters: the joint optimization of transmit power, bandwidth, and number of antennas is presented in Chapter 6, while the extension to advanced sleep modes, latency constraints, and dynamic scheduling is presented in Chapter 7. The power consumption modeling of BSs used in Paper C and D is discussed in Chapter 5.

1.2 Not Included in the Thesis

The following publications are not included in the thesis. They are conference versions of the journal Papers A,B and C respectively.

- **Optimizing Reconfigurable Intelligent Surfaces for Small Data Packets: A Subarray Approach.**
  Published in: *IEEE International Conference on Communications (ICC)*, Seoul, South Korea, pp. 2664–2669, 2022.
- **Control Signaling for Reconfigurable Intelligent Surfaces: How Many Bits are Needed?**
  Published in: *IEEE International Conference on Communications (ICC)*, Montreal, Canada, pp. 6856–6861, 2025.
- **Fundamentals of Energy-Efficient Wireless Links: Optimal Ratios and Scaling Behaviors.**
  Published in: *IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, Singapore, pp. 1–6, 2024.

1.3 Importance of Sustainability

As the world faces challenges of environmental conservation, economic stability, and human well-being, designing sustainable wireless networks is becoming increasingly important. Sustainability is a broad notion, and in this thesis we single out the three aspects that are most directly affected by the EE of a wireless link, namely environmental, financial, and human sustainability. Each is described in turn below.

1.3.1 Environmental Sustainability

Lowering the EC in wireless networks go hand in hand with a green sustainable future. Since a large portion of the world's energy is still generated from fossil fuels, lowering the EC leads to a decrease in carbon dioxide and other greenhouse gas emissions. This is crucial for mitigating climate change through a reduced carbon footprint, and it is the sustainability aspect with the clearest connection to the contributions of this thesis. If one can find an EE solution to a problem that can satisfy the QoS requirements, then such a solution will generally be better for the environment as well. An important observation is that

adding more hardware to a network is not by itself sufficient to make it more efficient. More antennas, wider bandwidths, and additional carriers all raise the passive PC of a site, so the efficiency gains only materialize if the use of that hardware is governed properly, by configuring it at the right operating point and by powering down the parts that are not needed. This thesis is concerned with precisely that governing problem. The advent of RIS can aid in making wireless networks more adaptable and environmentally friendly, aligning with the growing global emphasis on sustainability in technology.

1.3.2 Financial Sustainability

Reducing EC in wireless networks has a substantial economic impact, particularly from the perspective of mobile network operators (MNOs). When EE is improved, the most direct and immediate benefit is the reduction in operational costs. Energy bills constitute a significant portion of an MNO's ongoing expenses due to the need to power the network equipment, including BSs, data centers, and network infrastructure. By optimizing the EC, these costs can be reduced, which can lead to long-term financial savings. While there might be initial costs associated with upgrading equipment or implementing new technologies (such as RIS), the long-term payoff in terms of reduced energy bills is substantial (although one might also have to consider the maintenance cost of a more complex network). This makes the network more cost-effective to run over time. One can also argue that lowering the cost of access to wireless networks can help in providing a more fair access to connectivity for people around the world.

1.3.3 Human Sustainability

Increasing the EE in wireless networks can play a significant role in reducing electromagnetic (EM) radiation levels and the associated heating. This has implications for Specific Absorption Rate (SAR) values as well [15]. The SAR is defined as the rate at which energy is absorbed per unit mass by a human body when exposed to a radio frequency (RF) EM field and is defined as power absorbed per mass of tissue and has units of watts per kilogram (W/kg) [16].

When BSs or UEs, are designed to be more EE, they might transmit signals effectively at a lower power. This directly results in less EM radiation being emitted into the environment. Emerging technologies such as RIS utilized in conjunction with beamforming allow for more precise targeting of radio signals. This means that signals are directed specifically towards the intended user, rather than being broadcast in all directions, which reduces overall radiation levels and mitigates unnecessary exposure in areas where the signal is not required.

1.4 Research Objectives

The main objective of this thesis is to study how to increase the energy efficiency and manage the practical configuration overhead from both a base station's (BS's) and user equipment's (UE's) perspective. The approaches taken to enhance energy efficiency differ significantly between the two due to their distinct operational constraints. The UE is a highly power-constrained device operating on limited battery capacity, making the utilization of a passive component like a reconfigurable intelligent surface an ideal solution to improve channel conditions and directly reduce the UE's required active transmit power. However, realizing these gains introduces a critical system-level challenge: the BS must continuously compute and wirelessly transmit phase-shift configurations to the reconfigurable intelligent surface. Therefore, an essential focus of this thesis is designing efficient quantization codebooks to minimize this control signaling feedback overhead without sacrificing the achievable signal-to-noise ratio. In contrast, the BS relies on power-intensive, active multi-antenna arrays to serve the network. Consequently, to effectively improve energy efficiency at the BS side, the focus naturally shifts toward the joint optimization of its active and passive hardware parameters, alongside the implementation of advanced sleep modes to manage its large static power consumption.

In particular, we try to answer the following research questions (RQs):

RQ1: Can the total EC of a UE be reduced by using a RIS, if we account for the EC required to estimate the channel and configure it? How can the EC of the UE be reduced by optimizing pilot sequence length and transmit power while configuring the RIS according to the payload length?

RQ2: How can the wireless control signaling overhead required to feed back phase-shift configurations from the BS to the RIS be minimized while maintaining the SNR?

RQ3: For a BS, what is the optimal joint configuration of transmit power, bandwidth, and number of antennas to maximize EE? Furthermore, how do the sleep mode-specific transition times dictate this optimal hardware configuration?

RQ4: For a dual-band BS combining a coverage-oriented sub-6 GHz carrier with a capacity-oriented high band carrier, what operational regimes govern their joint optimal behavior? How should traffic be allocated between the two bands?

We address these questions under modeling assumptions that enable exact mathematical analysis. This contributes to the fundamental understanding of how to build energy-efficient wireless systems.

1.5 Research Methodology

The principal method used throughout this thesis is mathematical analysis and optimization. Each research question is turned into an optimization problem over a small number of physically meaningful variables, which is then treated analytically: we establish structural properties such as convexity or unimodality, derive closed-form expressions for the optimizers where the problem admits them, and design iterative algorithms where it does not. Numerical experiments are used alongside the analysis rather than in place of it. The work proceeds in four phases: conducting a literature review, identifying research gaps, formulating the problem mathematically, and subsequently answering the research question. The remainder of this section describes how these methods were applied in each of the four appended papers.

For RQ1, we were curious about understanding when it was worthwhile utilizing the RIS given the additional overhead it imposes upon the pilot transmission phase. The literature survey revealed a research gap in strategies for reducing the pilot overhead, and no clear understanding of how much power should be spent on pilot signals compared to data signals, with the case of a finite payload being particularly neglected. In Paper A, we therefore mathematically formulate an optimization problem and, through convexity analysis and numerical studies, investigate the pilot length, pilot power, and data power for different payload lengths and path losses in a system featuring an RIS.

For RQ2, the motivation behind Paper B was that the control signaling required to configure an RIS is itself an energy cost that is rarely accounted for. We approached this as a quantization problem, deriving how the available feedback bits should be distributed between the beam direction and the alignment with the static path, and bounding the SNR loss that a finite bit budget incurs.

For RQ3, the motivation behind Paper C was to dive deeper into the fundamentals to discover new relationships that can be used to increase the EE of wireless links. We mathematically formulate an optimization problem in the transmit power P, the bandwidth B, and the number of antennas M, constrained by upper bounds $(P_{\max}, B_{\max}, M_{\max})$,

$$\begin{aligned} \underset{P,B,M}{\text{maximize}} \quad & \text{EE} \\ \text{subject to} \quad & 0 < P \leq P_{\max}, \\ & 0 < B \leq B_{\max}, \\ & M \in \{1, \dots, M_{\max}\}, \end{aligned} \tag{1.1}$$

and subsequently extend it with sleep modes, QoS rate requirements, and latency budgets. We employed classical analysis and the Lambert W function to obtain closed-form solutions for the resulting transcendental equations. Alternating optimization algorithms were utilized to identify the maximum EE

points in a three-variable environment. By iteratively updating one variable while holding the others fixed, this method exploits the closed-form maximizer of each sub-problem to ensure monotonic convergence to a stationary point of the joint problem. The theoretical foundations for these optimization and analytical methods are established in literature [17, 18, 19, 20].

For RQ4, Paper D applies the same analytical framework to a dual-band BS. Here the solution is a partition of the traffic-load axis into regions, each with its own optimal configuration, together with a closed-form threshold that determines when the secondary band should be activated.

1.6 Methodological Justification

This thesis relies primarily on analytical mathematical optimization, with numerical simulation used to build understanding of the optimization results at practically relevant parameter values. The choice is motivated both by the nature of the research questions and by an identified gap in the literature. A substantial body of simulation-based work already exists on energy efficiency in wireless networks, yet the fundamental analytical characterization of optimal operating points, activation thresholds, and hardware configuration rules remains comparatively underexplored. This thesis aims to fill that gap. The research questions seek generalizable design rules (such as thresholds, optimal ratios, and activation conditions) which by their nature require closed-form characterization rather than scenario-specific evaluation. Simulation alone cannot establish global optimality or reveal how a threshold depends on hardware parameters. Experimental measurements might tie the conclusions to one specific hardware vendor's hardware. Several of the phenomena studied here such as the coupling between optimal transmit power and active antenna count, or the interaction between sleep-mode depth and latency constraints, can be observed in simulations but not explained by it. An analytical treatment is required to expose the underlying structural mechanisms and articulate them as design rules. The power consumption models considered here are in general smooth and differentiable, making Karush-Kuhn-Tucker (KKT)-based optimization directly applicable, and the exponential structure of the capacity expressions admits closed-form solutions through the Lambert W function.

1.7 Declaration of Use of Generative AI

During the preparation of this thesis and the appended publications, AI-assisted tools (large language models) were employed as productivity aids for code development, debugging, text editing, and analytical support. Paper A was completed without any AI assistance. For the remaining publications, Gemini 3.1 Pro was utilized for text drafting across Papers B, C, and D, as well as for mathematical analysis, assisting in the derivation of Theorem 3 in Paper C and Theorems 1 and 3 in Paper D. Notably, this model suggested the high-rate approximation that enabled the derivation of closed-form expressions in Paper D, and it was

partially used to generate the system model figure in Paper D. Claude Opus 5 was employed for text revision and language refinement. Claude Fable 5 was used to refine the dynamic scheduling framework in Section 7.7. While the author initially designed a baseline scheduling algorithm, the model-assisted refinement demonstrated an approximate 25% reduction in energy consumption and was therefore incorporated into the final thesis. All AI-generated suggestions, mathematical derivations, and algorithmic updates were thoroughly reviewed, verified, and validated numerically. AI tools were not used to generate research ideas, conduct literature reviews, or draw conclusions.

1.8 Societal Value of the Thesis

The continued growth of wireless data traffic, driven by the deployment of 5G and the upcoming transition to 6G, has made the energy footprint of mobile networks a pressing societal concern [8]. The radio access network already accounts for the majority of the electricity consumed by mobile operators [21], and a substantial share of the world's electricity is still generated from fossil fuels. Improvements in the energy efficiency of wireless links therefore translate, however indirectly, into reduced greenhouse-gas emissions, and any analytical understanding that helps operators run their existing infrastructure closer to its fundamental limits has a direct environmental benefit. This concern is increasingly reflected at the standards and operator level, with energy-efficiency targets now being defined by international bodies [12] and prioritized by mobile network operators [13].

The work in this thesis contributes to this direction by developing analytical foundations for energy-efficient operation rather than relying on incremental simulation studies. By deriving closed-form rules for how transmit power, bandwidth, antenna count, sleep-mode depth, and dual-band activity should be coordinated, the thesis provides design guidelines that hardware vendors and mobile network operators can apply across product generations, even as the underlying hardware-characterizing constants improve [9]. The proposed RIS subarray design and logarithmic-scaling feedback codebook reduce the configuration overhead that has so far been an issue for practical RIS deployment, while the dynamic sleep-mode scheduler shows that aggressive use of deep sleep states is compatible with the strict latency budgets of modern services.

Beyond the environmental dimension, more energy-efficient networks are also more economically sustainable: lower operating costs make it more feasible for operators to extend coverage to underserved areas, and lower transmit-power requirements reduce the electromagnetic exposure of users in the vicinity of BSs and handsets, which has direct implications for specific absorption rate values [15, 16]. In this sense, the contributions of this thesis support not only the climate goals associated with the telecommunications sector, but also the broader goal of providing reliable connectivity at a lower environmental and human cost.

1.9 Thesis Contributions

Regarding RQ1, we showed that there exists a unique energy-minimizing solution. We also analyzed what this solution looks like and concluded that for small payloads and when the direct path loss between the BS and UE is weak compared to the path loss via the RIS, the solution is using subarrays with many elements and low pilot power and vice versa. The optimal percentage of energy spent on pilot signaling is in the order of 10-40%. This means that it is optimal to let the pilot overhead constitute a significant portion of the EC because a good channel estimation is important to utilize the RIS to improve the EE.

Regarding RQ2, we uncovered that there was no clear understanding of how to co-optimize the power, bandwidth, and number of antennas for the current state-of-the-art PC models for EE. Rather than creating new models, we focused on analyzing the fundamentals and discovering new relationships that might not have been thought of yet. Within the confines of our model, we made an interesting discovery that at the global optimum the input transmit power is equal to the passive transceiver PC if the system is bandwidth limited. In addition, we developed an algorithm which converges to the global optimum. Our analytical approach lead us to novel insights regarding the optimal ratios between transmission power, the number of antennas, and bandwidth under both fixed constraints and unbounded solutions.

Regarding RQ3, we found that the joint optimization of transmit power, bandwidth, and number of antennas in a multi-antenna BS admits a simple rule: at the EE-optimal operating point, the radiated power equals the passive transceiver power consumption, so the optimal power-to-antenna ratio is dictated solely by the hardware-characterizing constants and is independent of the channel gain and the target rate. Building on this, we extended the framework to incorporate sleep modes and latency constraints. We showed that for absolute and deep sleep, the EE-maximizing strategy is a “rush-to-sleep” policy in which the BS transmits at the unconstrained optimal configuration and scales the activity factor linearly with the required rate, until a hardware-specific threshold is reached beyond which the BS is forced into a high-rate regime where sleep modes become unviable. Strict packet deadlines and finite wake-up transition times further constrain how deep a sleep state the BS can afford to enter. We translated these analytical insights into a dynamic event-driven scheduling algorithm that continuously balances the energy savings of deeper sleep against the energy penalty of compensating for wake-up delays.

Regarding RQ4, we mapped out the EE-optimal behavior of a dual-band BS combining a coverage-oriented sub-6 GHz primary carrier with a high-bandwidth, capacity-oriented sub-THz secondary carrier. By jointly optimizing the transmit power, number of active antennas, and sleep-mode activity factors of both bands, we identified four operational regions: primary-band duty-cycling, primary-band saturation, secondary-band duty-cycling, and dual-band saturation. We

derived an analytical threshold, expressed in closed form via the Lambert W function, that determines exactly when the secondary band should be awakened from sleep, and showed that activation is EE-optimal only when the power cost of forcing additional traffic through the bandwidth-limited primary band exceeds the static power penalty of bringing the secondary circuitry online. We also established that in the very high-rate asymptotic regime, the optimal traffic split between the two bands becomes independent of the channel conditions and is dictated entirely by the ratio of their bandwidths and power amplifier (PA) efficiencies.

1.10 Thesis Outline

The thesis is structured in the following manner. Chapter 2 provides the necessary background, covering the architectural design of BSs, an introduction to reconfigurable intelligent surfaces, and the channel and estimation models used throughout the thesis. Chapter 3 studies how to reduce the uplink energy consumption of a UE served by an RIS, examining how subarray size and pilot power allocation jointly determine the total energy cost of a short transmission. Chapter 4 turns to the control signaling required to convey the resulting configuration to the surface, presenting a codebook design whose feedback load scales logarithmically rather than linearly with the number of elements, together with extensions to 3D geometries and mobility. Chapter 5 develops the power consumption models that underpin the remainder of the thesis and derives the fundamental upper limits of energy efficiency that the later optimizations are measured against. Chapter 6 derives the optimal ratios and scaling behaviors for the joint configuration of transmit power, bandwidth, and number of antennas at a multi-antenna BS. Chapter 7 extends that framework to advanced sleep modes, latency constraints, and dynamic scheduling under bursty stochastic traffic. Chapter 8 lifts the analysis to a dual-band BS site, identifying the operational regions that govern the joint behavior of a sub-6 GHz carrier and a sub-THz carrier. Finally, Chapter 9 summarizes the main findings and discusses directions for future research.

2 Background

In this chapter, we describe the building blocks that are necessary to understand the contents of this thesis. We give an overview of EC reduction techniques, the basics of RIS, and the channel models used in our appended papers. The hardware-component PC modeling, sleep modes, and fundamental upper limits on EE are deferred to Chapter 5.

2.1 Energy Consumption Reduction Technologies

Historically, EE and EC have both increased simultaneously since the traffic grows so very rapidly. This means that while the EE as measured in [bit/Joule] is increasing, the total EC of the network as measured in [Joule] also does so. Many EC technologies work together to reduce the total EC in wireless networks. Some of these are investigated in this thesis while others constitute favorable research directions one might investigate which can work in tandem with those presented in this thesis.

- *Lean carrier*. In past generations of mobile networks (MN), meaning 4G and earlier, signals used for BS detection, system information, and channel estimation were continuously active or "broadcasted" very frequently, irrespective of the BS's engagement with UEs [22]. The point of a lean carrier is therefore to reduce the overhead in the signal transmission. Overhead refers to non-user-data parts of the signal, such as control information and signaling. By minimizing this, more of the network's capacity can be dedicated to transmitting actual user data. With less overhead, the system requires less power for processing and transmission of unnecessary reference signals (the cell-specific reference signal (CRS) in 4G was a specific villain). Lean carrier is particularly important for conserving energy for battery-powered devices.

- *Carrier shutdown*. By shutting down carriers (i.e. frequency bands) that are not actively being used for data transmission, significant energy savings can be achieved. This is because maintaining an active carrier, especially in high-frequency bands where the bandwidths are large, consumes considerable power considering no or little data is being transmitted. In [23] roughly 50% of energy is saved when the carrier is shut down in a network without load.
- *Bandwidth part (BWP)*. The BWP concept in 5G networks allows for the division of the overall carrier bandwidth into smaller parts, known as BWPs. Each BWP can be managed independently, allowing for more flexible and efficient use of the spectrum. For UEs, supporting a smaller BWP means that their radio hardware requirements are relaxed since The UE only needs to process a smaller bandwidth [24].
- *Symbol shutdown*. Symbol shutdown is aimed at optimizing EE by accounting for short-term fluctuations in traffic load within a cell. This allows for the swift deactivation of BS hardware components based on the current traffic conditions. This is achieved through selectively halting the transmission of certain symbols that may not be needed during periods of low data traffic. This leads to less transmitted power. Symbol shutdown is functioning within a time frame ranging from hundreds of microseconds to hundreds of milliseconds [9].
- *Sleep mode*. Energy savings increase with the number of network components turned off and the duration of their inactivity [25]. Modern wireless networks are designed to conserve energy by reducing PC for a transmitter or receiver that is not in active use. Different types of sleep modes include periodically waking up to check for paging messages as well as micro sleep during short periods of no scheduled traffic, which can be applied when a device is inactive for very short time intervals to conserve battery life. In deep sleep a significant portion of hardware is turned off or set in standby mode.
- *Selective hardware shutdown*. This involves turning off specific components or modules within the BS. One example is utilizing fewer antenna ports, meaning turning off parts of the transceiver hardware of the BS to reduce the circuit PC and computations required to process the advanced beamforming algorithms [26].

There is an inherent trade-off between EE and latency. One can increase the EE by placing the BS or the UE in sleep mode to conserve power. How long the device is sleeping does however impact the latency that is achieved.

Likewise, there is a trade-off between EE and coverage. One can easily conserve energy by turning off unused hardware that is not needed in low-load scenarios to reduce energy consumption. The issue is that in order to provide coverage in the network some reference signals such as the Synchronization Signal Block

(SSB) has to be transmitted still. Exactly which signals and carriers that can be turned off is not immediately obvious if the network functionality is to be maintained. This practical consideration is of great interest to MNO and BS hardware vendors alike.

Whether or not there is a trade-off between EE and data rate is a more complicated question. It depends on how many components one includes and the data rates, bandwidth available, and the considered model. We investigate this issue in our second paper.

Implementing selective hardware shutdown and sleep modes requires sophisticated control systems capable of monitoring usage patterns and making real-time decisions about which components to deactivate and when to enter which sleep mode without impacting the user experience negatively. We can see that despite several efforts to improve the hardware and software in BSs in the past few years, practical results still indicate that a BS with a low load consumes a similar amount of power as a BS experiencing a full load. The benefits of deep dormancy are mostly visible in zero-load scenarios when baseband processing and carriers can be fully turned off. In practical scenarios, max EE is achieved during full load as all the hardware is utilized to deliver the highest data rate possible while the increase in EC remains comparatively low.

2.2 Reconfigurable Intelligent Surfaces

RIS is a new emerging technology in wireless communications. These surfaces possess the unique capability to dynamically manipulate the reflected phase of incident EM waves with some precision. They are composed of numerous small, low-cost passive elements, each of which can independently manipulate EM waves through the phase of the reflection. The key feature of RIS is the reconfigurability. They can be programmed to dynamically change their properties in response to the wireless environment or network demands. By intelligently adapting the phase response to changing channel conditions or communication requirements, RIS can perform different tasks such as enhancing signal strength, mitigating interference, and increasing coverage.

2.2.1 Prospective Use Cases

Several use cases are envisioned for RIS. These include but are not limited to raising the received SNR, extending the coverage of the network, or increasing the EE. RIS can also help in capacity improvement, not just through an SNR boost but also by artificially improving the channel rank or through interference mitigation. Other possible applications include physical layer (PHY) security [27], RIS-assisted wireless power transfer (WPT) [28], non-orthogonal multiple access (NOMA) [29], along with localization [30] and sensing [31]. More applications can be found in [32].

2.2.2 Opportunities and Challenges

Several key challenges have to be tackled before RIS may be implemented in real networks. The first of these is how to provide an adequate performance gain while managing the increased pilot overhead. When utilizing an RIS, there has to be extra pilot resources spent on configuring it. The question is whether those resources spent outweigh the benefits since a large number of elements are needed in the RIS to provide a performance gain [33, 34]. This is an issue which we provide a solution strategy for in Paper A. Another challenge is how to design efficient RIS control signaling systems and algorithms that in real time can be transmitted from the BS to configure the RIS. At this point, RIS is a promising technology in its early stage of research for enabling us to improve the performance in the next generation of wireless systems.

Competing techniques such as relaying or small cells or distributed MIMO all rely on active transmission of signals which require PAs to operate, consuming substantial amounts of energy in the process. In contrast, RIS is a passive technology, meaning that it does not transmit any extra energy in itself, but rather it can reflect existing EM waves in the environment in desired directions to enhance, for instance, the SNR and coverage without increasing the total transmit power at the BS or the UE. Indeed, if the conditions for RIS-aided communication are beneficial, it might be possible to use less total transmit power to reach the target by relying on the reflections via the RIS compared to if the RIS had not been there. This makes the technology particularly attractive from an EE point of view since the only energy that is consumed is the one that is used in the internal electronics of the RIS itself along with the signaling and signal processing. Of course, integrating an RIS in a network also brings along an implicit energy cost at the BS side of the extra control signaling and signal processing that is required to operate the RIS. A study of how large an RIS has to be to compete with relaying systems can be found in [35].

The optimization of the phase shifts is non-trivial. Three different channels have to be learned to create optimal conditions for performance. The channel between the UE and the BS, the channel between the BS and the RIS, and the channel between the RIS and the UE. If multiple antennas are featured on the BS and the UE then the computational efforts that have to be spent to find the best set of phase shifts can be very substantial.

Ideally, the reflected signal from the RIS and the direct path should constructively interfere with each other, but finding such optimal phase shifts can prove time-consuming. Especially if one considers the uncertainty in the channel estimation. To optimize the system for EE, the efforts spent on channel estimation should be properly matched to the payload length and the relative importance of the RIS in the particular communication scenario. Utilizing the RIS gives a large performance boost when the contribution of the direct path is relatively small and in such situations it is enough to design the phase shifts, considering only the link between the UE and the RIS. We discuss this in Paper A.

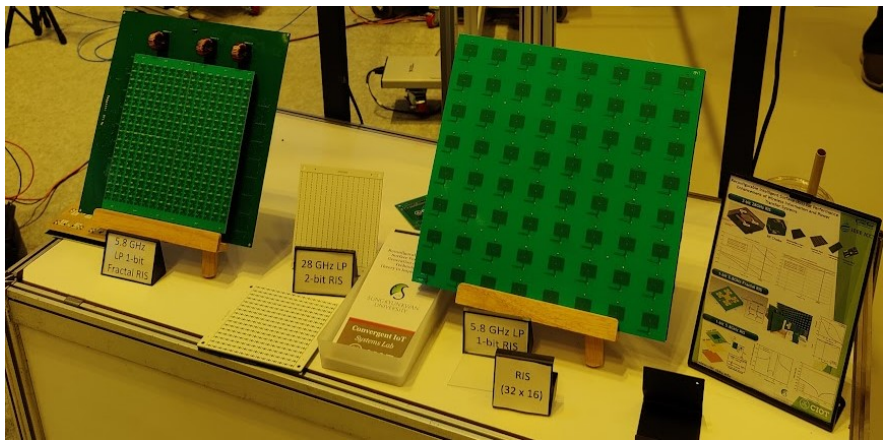


Figure 2.1: Examples of prototype RIS from Sungkyunkwan University in Seoul, one prototype is using operating at 5.8 GHz utilizing 512 elements with 1-bit phase shifts per element [43], and one is operating at 28 GHz utilizing 256 elements with 2-bit phase shifts per element.

2.2.3 RIS Examples

There are practical realizations of RIS in prototype form that provide the desired functionality, but their exact implementation can vary greatly between applications and different frequency bands. In typical scenarios relevant to frequencies in millimeter wave (mmW) bands down to low band frequencies it is common to utilize lumped electronic implementations such as described in [36]. These generally are based on switching or varactor diodes. These and other examples can be found in Table 1 in [37]. There have been several working prototypes developed implementing 1, 2, or even 3 bits. The number of bits refers to the resolution of the phase shift that each element in the RIS can achieve. Specifically, a b-bit resolution means that each RIS element can provide 2^b distinct phase shifts. Implementations of a 1-bit RIS can be found in [38, 39], 2-bit in [40], and 3-bit in [41]. There is also a possibility to manufacture RIS for optical frequencies or micro-electro-mechanical-system (MEMS) implementation of steerable mirrors [42]. An image of mid-band and mmW RISs is shown in Fig. 2.1.

2.3 Channel Models and Estimation

This thesis uses three channel models, each suited to a different deployment regime. A spatially uncorrelated Rayleigh fading model captures the UE-to-RIS link in the non-line-of-sight (NLoS)-dominated scenarios of Paper A, where many independent scatterers contribute paths of similar strength but uncorrelated phase. A deterministic LoS model with a per-element phase progression captures the BS-to-RIS link, which is typically engineered to be in LoS, and the BS-to-UE direct link in the open-environment scenario of Paper B and D. We

also include a brief treatment of how the channel can be estimated in RIS-aided systems, since the energy cost of this estimation is the central topic studied in Paper A.

2.3.1 Spatially Uncorrelated Rayleigh Fading

Spatially uncorrelated Rayleigh fading is used in Paper A as the channel model between a UE that is served by an RIS which is in NLoS.

Independent and identically distributed (i.i.d.) Rayleigh fading is a widely used mathematical model in wireless communication to characterize the random variations in signal strength due to environmental factors that cause multipath propagation. The model assumes that the fading amplitudes of different signal components are statistically independent but follow the same Rayleigh distribution. In this context, the Rayleigh fading characterizes the channel gain between the transmitter and receiver as a complex normally distributed random variable. This model is mostly accurate when numerous small-scale reflections and scatterings of similar power but with different phases contribute to the signal propagation.

The model is based on the assumption of independence of fading on different signal paths and the similar amplitude gains of these reflections. It is particularly relevant in scenarios where there is no dominant LoS path (in such cases Rician fading is more appropriate), and each multipath components contribute with similar strength as the others to the received signal. This often occurs in urban environments, indoor spaces, or areas with obstructions that cause signal reflections. This multipath behavior is supported by empirical sub-6 GHz measurements, which demonstrate that scattered signals arrive in distinct spatio-temporal clusters with significantly larger azimuth and delay spreads in urban versus suburban environments [44].

i.i.d. Rayleigh fading provides a convenient framework for analyzing and simulating wireless communication systems due to its statistical simplicity. However, its applicability is limited in scenarios where other propagation phenomena, such as shadowing or non-isotropic fading, play a significant role. In environments with strong line-of-sight components, or in situations where the channel characteristics deviate significantly from the assumptions of i.i.d. Rayleigh fading, alternative models may be more appropriate for accurate representation of the wireless channel.

Despite its limitations, we have used it in Paper A to analyze the performance of RIS-aided systems. This can be motivated by the fact that the UE is likely to not always be in LoS to its serving RIS (or BS) and it can therefore be used as a good estimate of a wireless channel.

We here derive the channel model for Rayleigh fading. Assume that the UE is in NLoS with the RIS and transmits a narrowband signal via the RIS composed

of M elements, modeled as a uniform linear array (ULA).

By modeling a rich scattering environment with N independent paths, the channel to each element m is uncorrelated across paths. Throughout this thesis we work with the equivalent baseband (complex envelope) representation of the channel; the carrier frequency f_c is suppressed. The continuous-time channel impulse response from the source to element m can be modeled as

$$h_m(t) = \sum_{n=1}^{N} \sqrt{r_{m,n}} e^{-j\psi_{m,n}(t)} \delta(t - T_{m,n}). \tag{2.1}$$

We have that $\sqrt{r_{m,n}}$ is the channel gain, $T_{m,n}$ is the time delay of each path, and $\psi_{m,n}$ the phase rotation of each path. If the magnitude of each path is of equal strength but includes a uniform independent phase rotation, the channel gain of a sampled discrete-time version can be modeled as a circularly symmetric complex Gaussian [45] such that

$$h_m \sim \mathcal{N}_{\mathcal{C}}(0, \beta) \tag{2.2}$$

where β is the variance and $|h_m|$ is said to be Rayleigh distributed.

The Rayleigh distribution has a cumulative distribution function (CDF) given by:

$$F(x; \beta) = 1 - e^{-x^2/\beta}. \tag{2.3}$$

It is interesting to see how many paths N are needed to make the approximation valid. In Fig. 2.2 we have plotted the CDF of $|h_m(t)|$ for different numbers of paths N. Through checking how the curves match, it is evident that the approximation is struggling if $N \leq 3$ but quite good for $N \geq 5$.

If the Rayleigh distribution from the UE to the RIS is combined with a LoS path between the BS and RIS to make a concatenated channel, then the total phase shift between the UE and BS will also be uniformly randomly distributed. To optimally configure the RIS, the pilot length will have to be matched to the number of elements. There is therefore no *a priori* structure in the cascaded channel that the BS could exploit to configure the RIS without first measuring it by transmitting pilot symbols. We defer the discussion of how many pilot symbols are required for this measurement to Section 2.3.4 below

2.3.2 LoS Channel Model

In case there is LoS between the UE and the BS or RIS, the channel is modeled differently. As in Paper A, we assume that the BS and the RIS are in LoS with

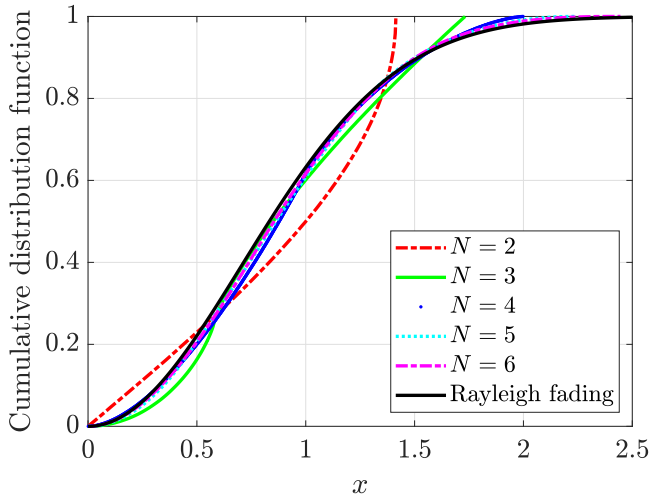


Figure 2.2: The CDF $\mathbb{P}\{|h_m| \leq x\}$ of $|h_m|$ with Rayleigh fading compared to $N = 2$ to $N = 6$ paths in (2.1) with constant gain and uniformly distributed phases.

one another. Since we have LoS paths from the RIS to the BS, the distance to each element is roughly the same from an amplitude point of view, so all elements share a common gain $\sqrt{\alpha}$. However, the small differences in path length across the array induce different propagation delays. Under the standard narrowband assumption, where the signal bandwidth is much smaller than the carrier frequency ($B \ll f_c$), these per-element delay differences $\Delta T_{m,n} = (d_m - d_n)/c$ are too short to resolve in the time domain but produce a measurable per-element phase shift $\varphi_m = 2\pi f_c \Delta T_{m,n} = 2\pi(d_m - d_n)/\lambda$. The time delay $\delta(t - T_{m,n})$ that appeared in the multi-path model (2.1) therefore collapses to a single complex coefficient per element in the narrowband regime, and the channel between the BS and RIS element m can be written as

$$h_m = \sqrt{\alpha} e^{-j\varphi_m} \tag{2.4}$$

where φ_m is the phase shift of element m w.r.t. the direct path, which is assumed to be the reference phase, i.e. $\varphi_1 = 0$. The phase difference depends on the path length difference between the elements and is given as:

$$\varphi_m = 2\pi(d_m - d_1)/\lambda \tag{2.5}$$

where d_m is the distance to element m, and λ is the wavelength.

2.3.3 RIS-assisted Channel Modeling

The end-to-end channel response between the single antenna UE and single antenna BS is

$$g = \sqrt{\rho} + \sum_{m=1}^{M} e^{-jc_m} \sqrt{\alpha} e^{j\phi_m} \sqrt{\beta}. \tag{2.6}$$

Where $\sqrt{\rho}$ is the gain of the direct path between the BS and the UE, $\sqrt{\alpha}$ is the gain between the BS and the RIS and $\sqrt{\beta}$ is the gain between the RIS and the UE. Without loss of generality, we choose the direct path as the phase reference. The phase shift of the concatenated channel between the UE and the BS via the RIS is denoted ϕ_m When the channel in (2.6) is used for data transmission, the SNR is

$$\text{SNR} = \frac{P_{\text{tx}}}{\sigma^2} |g|^2 . \tag{2.7}$$

For the configuration $c_1, \dots, c_M$ over the M elements, the SNR is maximized by $c_m = \phi_m$, which gives the maximum possible SNR as the optimized configuration

$$\overline{\text{SNR}} = \max_{c_1,\dots,c_M} \text{SNR}(c_1, \dots, c_M) = \frac{P_{\text{tx}}}{\sigma^2} \left(\sqrt{\rho} + M\sqrt{\alpha\beta} \right)^2 . \tag{2.8}$$

We can see that the end-to-end channel gain behaves as

$$|g|^2 = \left(\sqrt{\rho} + M\sqrt{\alpha\beta} \right)^2 \approx \begin{cases} M^2\alpha\beta & \text{if the direct path is weak,} \\ \rho & \text{if the RIS paths are weak.} \end{cases} \tag{2.9}$$

The contribution from the elements to (2.9) is M^2 when the dominant contribution to (2.9) comes from the RIS elements, which is the situation in which the RIS is the most useful. There is a contributing factor M from the number of elements which increases the received signal strength and another contribution M from the beamforming gain if the phase shifts are perfectly configured. The conclusion is that physically large RIS are significantly better than small RIS because of the quadratic scaling [46].

2.3.4 Channel Estimation

Due to the stationarity, the channel between the RIS and the BS is the easiest to learn, since both of them can be assumed to be stationary, deployed in LoS of one another. In deployments where significant non-stationary scatterers exist near the BS or RIS (vehicles, foliage in wind, temporary obstructions,

etc.) the BS-RIS channel inherits time-varying components that this assumption neglects. In such cases the BS-RIS channel must be re-estimated periodically, and the methods of Section 2.3.4 apply but with shorter validity windows. Throughout this thesis we assume the BS-RIS link is sufficiently stationary that its CSI can be treated as known. This channel is also unaffected by the mobility changes in the environment. It is therefore assumed that this channel is known or at least can be estimated in such detail that it essentially corresponds to perfect CSI. As we have established, due to the great number of elements that can be present in an RIS, the phase shifts have to be configured according to the parameters of the channel. The channel between the BS and the UE is learned through various methods, such as the transmission of uplink (UL) pilot signals or downlink (DL) feedback schemes. However, the estimation can take up a great deal of resources. There are two methods of estimation, transmission of pilot signals (also known as preambles, reference signals, etc.) either in the DL or in the UL. In 5G standards, the channel between the UE and the BS can be estimated in various configurations either via the Channel State Information Reference Signal (CSI-RS) in the DL or through an UL transmission from the UE to the BS through the Sounding Reference Signal (SRS). Each method has its own set of advantages and disadvantages. In this thesis, we will focus on UL transmission methods. It is worth noting however that in frequency division duplex (FDD) systems, channel reciprocity does not hold, therefore in FDD systems DL pilots are the only option for DL channel estimation.

To compute the RIS configuration that maximizes the SNR in (2.8), we need to learn the current channel realization. We collect the unknown channel coefficients into a single vector $\mathbf{h} \in \mathbb{C}^{M+1}$ whose first entry is the direct BS-UE path and whose remaining M entries are the cascaded BS-RIS-UE paths through each individual element. Let

$$\mathbf{h} = \begin{bmatrix} \sqrt{\rho} & \sqrt{\alpha\beta}e^{-j\phi_1} & \dots & \sqrt{\alpha\beta}e^{-j\phi_M} \end{bmatrix}^{\mathrm{T}} \in \mathbb{C}^{M+1} \tag{2.10}$$

denote the vector containing these channel coefficients. Since there are $M+1$ unknown coefficients, we assume the UE sends a predefined pilot sequence of length $M+1$ in order to obtain sufficient degrees-of-freedom to estimate the individual coefficients in $\mathbf{h}$. Without loss of generality, the pilot sequence is constant $\sqrt{P_{\text{pilot}}}[1 \ \dots \ 1]^{\mathrm{T}} \in \mathbb{C}^{M+1}$, where P_{pilot} is the transmit power. During the pilot transmission, the RIS changes its configuration according to a structured deterministic phase-shift pattern. Let $\boldsymbol{\phi} \in \mathbb{C}^{M+1}$ be the vector with 1 as its first entry (representing the direct path) and the configured phase-shifts $e^{-j\phi_m}$ for elements $m = 1, \dots, M$ during the pilot transmission of symbol $i = 1, \dots, M$. This vector is given as

$$\boldsymbol{\phi}_i = \begin{bmatrix} 1 & e^{-j\phi_{i,1}} & \dots & e^{-j\phi_{i,M}} \end{bmatrix}^{\mathrm{T}}. \tag{2.11}$$

Then, the received signal at the BS is given in vector form as

$$\mathbf{y}_{\text{pilot}} = \sqrt{P_{\text{pilot}}} \underbrace{\begin{bmatrix} \boldsymbol{\phi}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{\phi}_{M+1}^{\mathrm{T}} \end{bmatrix}}_{\triangleq \Phi} \mathbf{h} + \mathbf{w}_{\text{pilot}} \tag{2.12}$$

where $\mathbf{w}_{\text{pilot}} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{o}, \sigma^2 \mathbf{I}_{M+1})$ is the thermal receiver noise. If we select the matrix Φ as any scaled unitary matrix (e.g., a discrete Fourier transform matrix), then $\Phi^H \Phi = (M+1)\mathbf{I}_{M+1}$ and the BS can estimate the individual paths as

$$\frac{\Phi^{\mathrm{H}} \mathbf{y}_{\text{pilot}}}{\sqrt{M+1}} = \sqrt{(N+1)P_{\text{pilot}}}\mathbf{h} + \frac{\Phi^{\mathrm{H}} \mathbf{w}_{\text{pilot}}}{\sqrt{M+1}} \tag{2.13}$$

where the effective noise component has the same distribution as $\mathbf{w}_{\text{pilot}}$ since $\Phi^{\mathrm{H}}/\sqrt{M+1}$ is a unitary matrix. Recalling the entries of the channel vector $\mathbf{h}$ from (2.10), we have different pilot SNR for the BS-UE direct path and the paths through the M elements of RIS. Using the channel statistics derived before, the pilot SNR for the direct channel and the cascaded RIS channels are given as

$$\text{SNR}_{\text{pilot}} = \begin{cases} \frac{(M+1)P_{\text{pilot}}\rho}{\sigma^2} & \text{for the direct channel,} \\ \frac{(M+1)P_{\text{pilot}}\alpha\beta}{\sigma^2} & \text{for the RIS channels.} \end{cases} \tag{2.14}$$

In our design problem, we will focus on the pilot SNR of the RIS channels that is given in the second line of (2.14).

2.3.5 Required Pilot SNR to Achieve Maximal Data SNR

When transmitting the pilot signals, they will be received with an SNR value of $\text{SNR}_{\text{pilot}}$ in (2.14) depending on the allocated power. If its value is not high enough, it will lead to the BS not being able to adequately compute the optimal phase shifts needed to reach the maximum data SNR in (2.8). The performance loss is defined as the ratio between the achieved SNR in (2.7) with the phase-shifts that are adjusted according to the channel estimates and the maximum in (2.8).

We used (2.13) to estimate the phase shifts when calculating the ratio $\text{SNR}/\overline{\text{SNR}}$ for different $\text{SNR}_{\text{pilot}}$ values. The power that is allocated to the pilot symbols increases the achievable data SNR albeit at the cost of additional EC. However, achieving the target SNR in the pilot transmission is only possible if the channel gains are known beforehand.

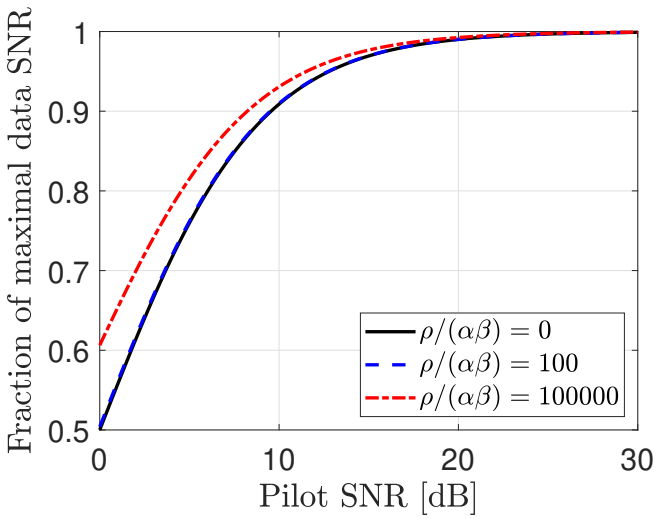


Figure 2.3: Relative performance loss in data SNR for $M = 1024$ individually configured elements as a function of pilot SNR as defined in the second line of (2.14).

In Fig. 2.3 we have simulated the fraction of maximal achievable data SNR for $M = 1024$ elements and varying path loss values. If the direct path is weak, (in the sense that $\rho/(\alpha\beta)$ is 100 or below), then we need a pilot SNR of 20 dB to get close to the perfect CSI case.

The key point here is that the pilot overhead is a non-negligible contribution to the EC and how much power should be allocated to the pilot signals is an important aspect of EE optimization.

3 Reconfigurable Intelligent Surfaces for Reducing Uplink Energy Consumption

The individual wireless links might not be capacity-constrained in 6G wireless technology, thanks to enormous bandwidths, but rather limited by EE. Most wireless communication applications generate intermittent traffic [47] which motivates the study of adapting the RIS configuration to payloads of different lengths. We define the *payload length* L as the number of data symbols the UE needs to transmit in a single transmission opportunity, distinct from the pilot length used for channel estimation. Practical payloads span several orders of magnitude: a few tens of symbols for Internet of things (IoT) control messages or short status updates, to thousands of symbols or more for sustained downloads. The relative weight of pilot overhead and data transmission in the total energy budget shifts dramatically across this range, and a configuration that is energy-optimal for short payloads is generally suboptimal for long ones. One of the main challenges with RIS is that the large number of elements requires a great deal of resources to be spent on channel estimation, an overhead that is particularly punitive when the payload itself is short.

3.1 Research Question

The main research questions are listed below:

- Can the EC be reduced by shortening the pilot sequences through the utilization of subarrays or reducing their power?
- Should the RIS be configured differently depending on the payload length?
- How are deployment factors, such as hardware impairments, imperfect

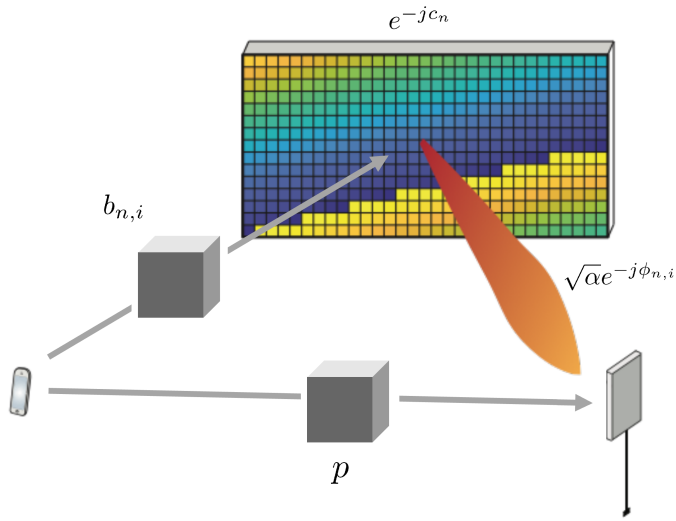


Figure 3.1: We consider a setup where a UE transmits to a BS through a direct path p and a controllable RIS with M elements. Two of the paths are in NLoS and one in LoS. The colors represent the phase shifts of the RIS elements.

CSI, and propagation conditions, impacting the answers to these questions?

3.2 Related Works

The energy consumption of RIS-aided systems has been studied from several angles in the literature. Closest to our work, the RIS-aided ergodic capacity of fading channels was analyzed in [34, 48], where the number of reflecting elements was treated as a design variable and optimized for spectral efficiency. These contributions establish that the EE of an RIS-aided link is non-monotone in the number of elements: more elements increase the array gain, but also inflate the pilot overhead required to configure them. Neither work, however, considers payloads of finite length. In Paper A we show that, when the payload is short, grouping elements into subarrays is a more energy-efficient strategy for reducing the configuration dimensionality than turning elements off altogether, since it preserves the array gain while shortening the pilot sequence.

Several authors have explored ways to reduce the pilot overhead by exploiting structure in the channel. Methods that exploit channel sparsity, low-rank structure, or known geometric priors can reduce the required pilot length below the per-element baseline; the maximum-likelihood estimator developed in [49] is a representative example for LoS-dominant channels. The grouping of RIS elements into subarrays sharing a common phase shift was proposed for the orthogonal frequency division multiplexing (OFDM) setting in [50], primarily as a means to reduce the dimensionality of the channel-estimation problem rather

than as an energy-minimization device.

Discrete phase-shift hardware introduces additional SNR loss compared to the ideal continuous-phase case, which has been characterized in [51, 52]. We incorporate this loss into our analysis through the bound $\mathrm{SNR}_{\mathrm{loss}} \leq \mathrm{sinc}^2(2^{-b})$ derived later in this chapter.

Compared to these prior works, the contribution of Paper A is to make the pilot–data energy trade-off the central object of analysis. We consider a finite payload of L symbols, formulate the joint optimization of subarray size and pilot power as an energy-minimization problem in (3.5), and characterize how the optimal configuration depends on the payload length and the relative strength of the direct and RIS-assisted paths. The explicit pilot-vs-data energy accounting has not been treated analytically in the prior literature.

3.3 Contributions

In Paper A, we formulate and solve an optimization problem regarding when it is worth combining multiple phase shifts into a single value versus how much power to distribute to the pilot versus data signals. As it turns out, closed-form expressions can be extracted for certain targets in both pilot and data SNR and later utilized in the optimization.

3.4 System model

In Paper A we consider, as described in Chapter 2, i.i.d. Rayleigh fading and the implications of the channel estimation and discrete phase shifts together with incorporating subarrays, in great detail in this scenario.

We consider a system where a single-antenna UE transmits to a BS (possibly equipped with multiple antennas). There is an RIS in the vicinity of the BS and UE that can be utilized to improve the channel conditions. We assume that the BS configures and interacts with the RIS.

3.4.1 Subarrays

A practical obstacle in deploying RIS-aided systems is the additional pilot signaling overhead required to identify the preferred RIS configuration [33]. The subarray approach addresses this by grouping several adjacent RIS elements into a *subarray* that is configured with a single common phase shift, reducing the effective number of independent control degrees-of-freedom from M to $N \leq M$. This approach has direct precedent in massive MIMO BS subarrays where antennas are controlled together, something which also has garnered interest in the field of RIS [50].

Subarrays reduce three distinct overheads simultaneously, all driven by the reduction from M to N degrees-of-freedom:

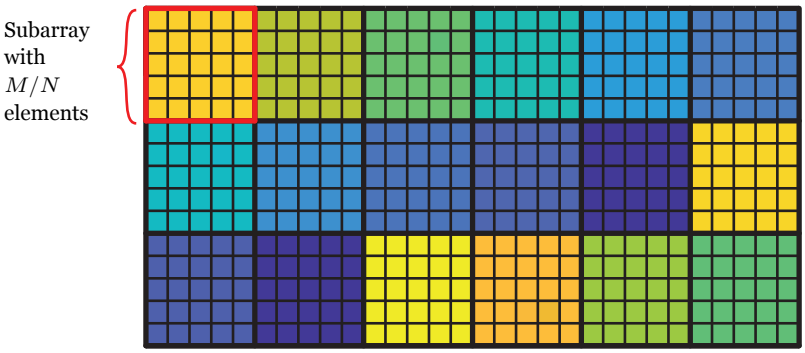


Figure 3.2: An example setup with an RIS consisting of $M = 450$ elements that are divided into $N = 18$ subarrays, each with $M/N = 25$ elements. This figure is reproduced from Paper A [1].

- **Pilot length.** Estimating the cascaded channel coefficient associated with each independently controllable subarray requires one pilot symbol per subarray (plus one for the direct path), so the pilot sequence shortens from $M + 1$ symbols to $N + 1$ symbols. This is the dominant saving in the energy analysis of Paper A.
- **Control signaling.** Feeding back the chosen configuration from the BS to the RIS requires transmitting a phase index per controllable element. With subarrays, the feedback payload scales with N rather than M, easing the load on the wireless control channel that we study in detail in Chapter 4.
- **Hardware control complexity.** Each independently controllable phase shift requires its own bias-control line and varactor (or PIN-diode) drive circuit. Driving N independent control voltages instead of M reduces the routing complexity, the number of digital-to-analog conversion stages on the RIS controller, and the computational cost of the algorithms that map estimated channels to phase configurations.

The trade-off is that within a subarray all elements share a common phase shift, so the achievable beamforming gain is reduced and doubling the elements per subarray no longer doubles the contribution to the received signal strength. The choice of N therefore balances configuration overhead against array gain.

We make a thorough analysis of subarrays in Paper A. An example is shown in Fig. 3.2 in which the different colors indicate the phase-shift values.

3.4.2 Channel Model

When considering the UL system model (with a single antenna on each side) between a UE and a BS, the UE transmits the data signal x with power P_{data},

i.e., $\mathbb{E}\{|x|^2\} = P_{\text{data}}$. The received signal at the BS is

$$y = gx + w, \tag{3.1}$$

where $w \sim \mathcal{N}_{\mathbb{C}}(0, \sigma^2)$ is the complex Gaussian thermal receiver noise and g is the end-to-end channel response which is equal to

$$g = p + \sum_{n=1}^{N} e^{-jc_n} \underbrace{\sum_{i=1}^{M/N} \sqrt{\alpha} e^{-j\phi_{n,i}} b_{n,i}}_{=Z_n}, \tag{3.2}$$

Because of the i.i.d. Rayleigh fading we have $p \sim \mathcal{N}_{\mathbb{C}}(0, \rho)$, and $b_{n,i} \sim \mathcal{N}_{\mathbb{C}}(0, \beta)$. We define $Z_n = \sum_{i=1}^{M/N} \sqrt{\alpha} e^{-j\phi_{n,i}} b_{n,i}$ as the concatenated channel of subarray n. We note that $Z_n \sim \mathcal{N}_{\mathbb{C}}\left(0, \frac{\alpha\beta M}{N}\right)$ and thus $|Z_n| \sim \text{Rayleigh}\left(\sqrt{\frac{\alpha\beta M}{2N}}\right)$.

3.4.3 Discrete Phase Shifts

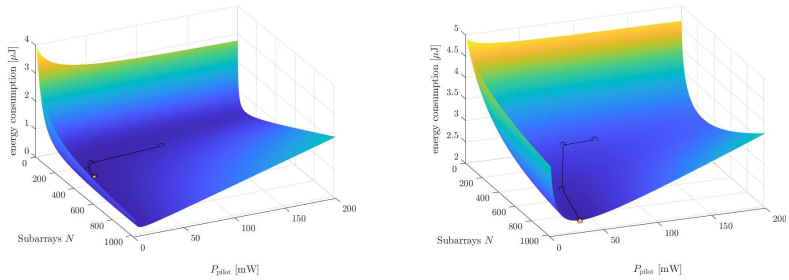
The RIS PIN-diodes are operated by a set number of control signals in the form of bits. The voltage applied induces a different phase shift which leads to the induced phase shift. More bits of control information (along with the hardware capable of supporting it) lead to a greater variety of phase shifts available, and thus greater performance but also increases the complexity and potentially the cost of the RIS design. In Paper A, we derive the relationship that the SNR loss incurred from discrete phase shifts is limited to

$$\text{SNR}_{\text{loss}} \leq \text{sinc}^2\left(\frac{1}{2^b}\right) \tag{3.3}$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$ is the sinc-function, b is the number of bits of control information, and 2^b is the resulting number of quantized phase shift states, which are assumed to be uniformly distributed across the unit circle in the complex plane. Note that $\text{sinc}^2(1/2) = 3.922\,\text{dB}$ for the 1-bit quantized RIS represents an upper bound on the highest possible SNR loss in logarithmic scale that may occur from using discrete phase shifts instead of the ideal case which employs continuous phase shifts [51]. An analysis of how the data rate is degraded by discrete phase shifts is found in [52].

3.5 Optimization Problem Formulation

We want to minimize the EC at the UE while ensuring successful decoding at the BS by meeting the SNR target. The EC $E(N, P_{\text{pilot}})$ of transmitting the $N + 1$ pilots during channel estimation (with N subarrays) and the L data symbols is



(a) (b)

Figure 3.3: Energy Consumption $E(N, P_{\text{pilot}})$ in a scenario with imperfect CSI when transmitting $L = 200$ symbols and the direct path is weak ($\rho = -110$ dB) in (a) and when transmitting $L = 10\,000$ symbols and the direct path is strong ($\rho = -90$ dB) in (b). The results obtained by the numerical optimization based on steepest descent are shown in black for 5, 10, and 15 iterations.

$$E(N, P_{\text{pilot}}) = (N+1)\frac{P_{\text{pilot}}}{B} + L\frac{P_{\text{data}}}{B} + (L+N+1)\frac{P_{\text{circuit}}}{B}, \tag{3.4}$$

where B is the symbol rate measured in symbol/s.

We solve the following optimization problem:

$$\begin{aligned} &\underset{N, P_{\text{pilot}}}{\text{minimize}} \quad E(N, P_{\text{pilot}}) \\ &\text{subject to} \quad N \in \{1, \dots, M\}, \quad \frac{M}{N} \in \mathbb{Z}, \quad P_{\text{pilot}} \geq 0. \end{aligned} \tag{3.5}$$

The optimization variables are the subarray size N and pilot power P_{pilot}. The required power in the data channel P_{data} is a function of them. We assumed that all other parameters were constants. We employed a gradient steepest-descent algorithm, which due to the smoothness of the function converged to the global optimum within a few iterations.

3.6 Results

3.6.1 Solving the Optimization Problem

The results in Fig. 3.3 paint an intuitive and quantifiable picture of the EC. Less power should be spent on pilot overhead in situations where the RIS is less useful i.e., either its channel gain is relatively low compared to the direct path, or the payload that has to be transmitted is small enough that it is unwarranted to spend significant efforts to transmit long and high-power pilot symbols.

3.6.2 Payload Length Impact

Since practical systems will transmit payloads of variable length L depending on the UE's traffic, different solutions to (3.5) are needed by different UEs and at different times. Hence, it is important to determine how the optimal energy-consumption solution parameters $(N^\star, P^\star_{\text{pilot}})$ behave together. We have already established that increasing either will also increase the received SNR at the cost of higher energy consumption. For a varying payload length, the energy-consumption optimization problem (3.5) was solved. The optimal solutions are shown in Figs. 3.4(a) and (c). Also shown is the required data power P_{data} in (b) and pilot energy in (d). The total percentage of energy spent on pilot signaling is shown in (e). As the payload L becomes larger it is generally worth it to spend extra energy to configure the RIS with fewer elements per subarray and more pilot power.

3.7 Summary

For small payloads, and when the direct path loss between the BS and UE is weak compared to the path loss via the RIS, the energy-minimizing solution uses subarrays with many elements and low pilot power; the opposite is true for long payloads and stronger direct paths. The transmitted pilot power levels of 10 to 40 mW in Fig. 3.4(c) translate to pilot SNR levels of approximately -5 to 0 dB per subarray. This means that, from an EC perspective, a low-resolution estimation of the channel to each individual RIS element is generally preferable to a high-resolution estimation of a larger subarray. The optimal percentage of energy spent on pilot signaling is in the order of $10{-}40\%$.

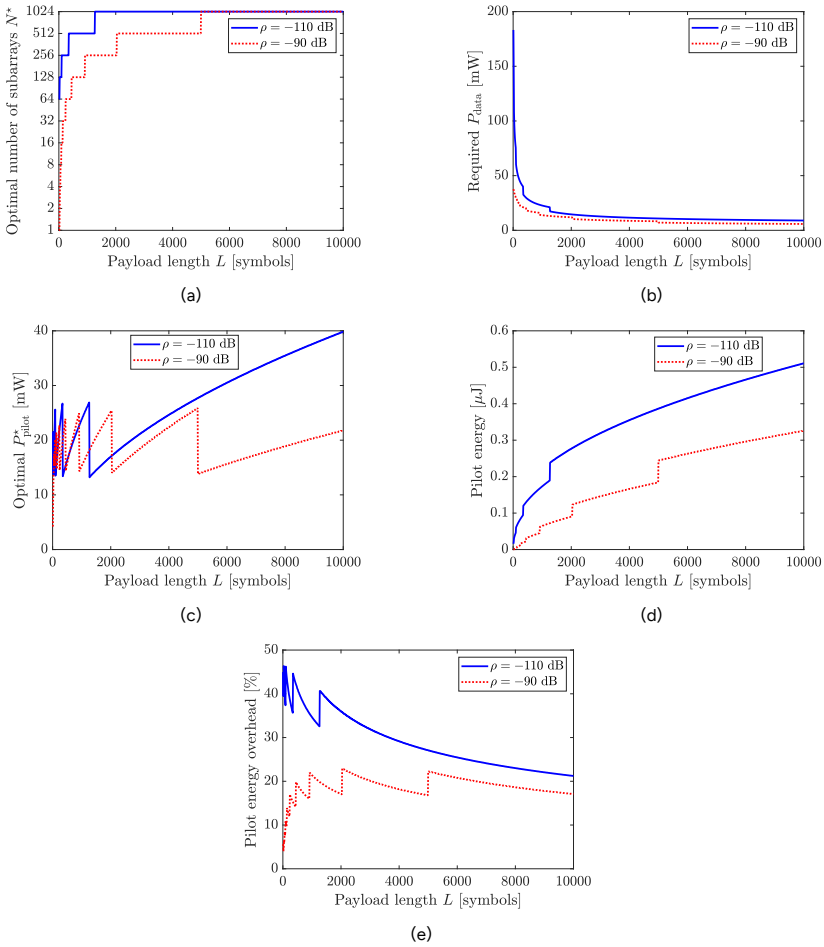


Figure 3.4: The optimal energy consumption minimizing subarray size (a) and pilot power (c), required data power (b), total pilot energy (d), and the percentage of energy spent on the pilot signaling (e), for different payload lengths are shown for a weak ($\rho = -110$ dB) and strong ($\rho = -90$ dB) direct path between the UE and BS.

4 Minimal Feedback Control Signaling for Reconfigurable Intelligent Surfaces

The previous chapter focused on the uplink energy consumption of an RIS-aided UE and on the pilot overhead required to estimate the cascaded channel. Even when the BS has perfect channel knowledge and can compute the optimal RIS phase-shift configuration, implementing this configuration requires the BS to convey it to the RIS over a wireless control channel. The main feature that motivates an RIS over an additional BS is precisely that it has no wired backhaul, so this control link must be wireless and consume as few radio resources as possible. In practical systems such as 5G and beyond, control channels either have limited dedicated resources or share resources with the data channel [53]. A naive element-wise feedback approach, where each RIS element receives its own quantized phase shift, scales linearly with the number of elements, which becomes prohibitively expensive for large surfaces. This chapter addresses the design of a feedback codebook that achieves logarithmic scaling instead, while still guaranteeing minimal SNR loss.

4.1 Research Question

The main research questions addressed in this chapter are:

- How does the size of the RIS configuration feedback message, measured in bits, affect the SNR compared to an unquantized ideal configuration?
- What are the performance trade-offs between codebook-based and element-wise feedback schemes, and how does the channel model affect them?
- How does the required feedback load scale with more complex geometries, such as 3D uniform planar arrays (UPAs), and with a non-zero

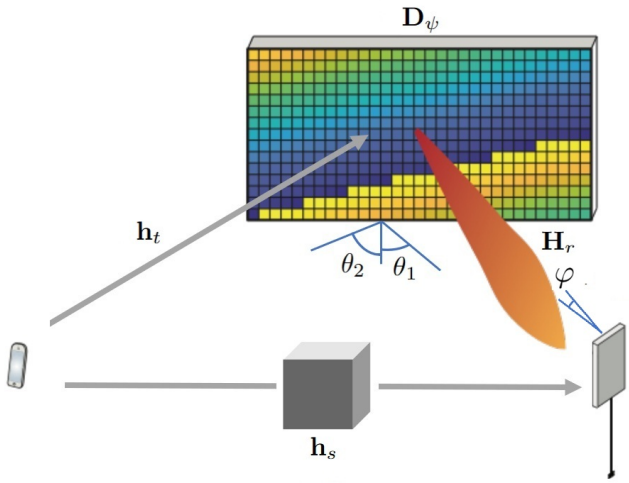


Figure 4.1: System model of Paper B. A UE transmits to a multi-antenna BS through a partially obstructed static path $\mathbf{h}_s$ and a controllable RIS with N elements. The channel from the BS to the RIS is in LoS at angle θ_1 at the RIS, and the channel from the RIS to the UE is in LoS at angle θ_2. The BS receives the signal from the RIS at angle φ. The control channel is used to feed back the phase-shift configuration to the RIS using t bits.

static path between the UE and the BS?

- How often should the RIS configuration be updated for mobile UEs, and can it be managed by low-overhead differential methods?

4.2 Related Works

Much of the prior literature on RIS-aided communication has focused on estimating the cascaded channel at the BS [54] and using the resulting estimates to compute the preferred RIS configuration; see [55] for a review of fundamental algorithms and results. These efforts aim to minimize the overhead associated with channel estimation, but typically assume the resulting configuration can then be transmitted to the RIS without consuming radio resources. More recent works have begun to address the control signaling itself. A neural network was designed in [56] to jointly optimize the BS precoder and the RIS configuration in a way that reduces information exchange. The impact of RIS control operations based on beam sweeping was evaluated in [57] and is the closest prior work, while [58] considered the joint overhead of CSI acquisition and configuration feedback. Codebook-based solutions specifically have gained traction [59, 60] due to their robustness and feasibility in field trials [61], and an environment-aware codebook design that uses statistical CSI to construct a reflection-coefficient codebook robust to channel-estimation errors was proposed in [60]. However, where these earlier works focus on constructing codebooks to acquire CSI with reasonable overhead, Paper B instead eval-

uates the performance with known CSI and determines the minimal feedback load required to update the RIS configuration. A bit-level analysis of the actual number of feedback bits required to achieve a certain SNR was missing from the prior literature, and this is the gap that Paper B fills.

4.3 Contributions

Paper B makes the following main contributions:

- A novel codebook design for LoS scenarios that reduces the required number of feedback bits from linear to logarithmic scaling with the number of RIS elements, while ensuring a bounded SNR loss.
- An extension of the codebook design that aligns the RIS-reflected beam with the static BS–UE path, and that supports 3D propagation environments and UPA-shaped RISs through a joint quantization of azimuth and elevation angles on a non-uniform circular grid.
- A low-overhead differential feedback scheme of only a few bits per update for mobility scenarios, together with the observation that beam-direction updates and static-path-alignment updates have very different temporal requirements.
- A combined SNR-loss analysis that accounts simultaneously for the codebook quantization and for the discrete phase-shift hardware imposed by practical RIS implementations.

4.4 System Model

We consider the uplink illustrated in Fig. 4.1, where a single-antenna UE transmits to a BS arranged as a ULA with K antennas, aided by an RIS that is initially modeled as a ULA with N elements. The end-to-end channel vector $\mathbf{h} \in \mathbb{C}^K$ is expressed as

$$\mathbf{h} = \mathbf{h}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{h}_t, \tag{4.1}$$

where $\mathbf{h}_s \in \mathbb{C}^K$ is the static path between the UE and the BS, $\mathbf{H}_r \in \mathbb{C}^{K\times N}$ is the channel from the RIS to the BS, $\mathbf{h}_t \in \mathbb{C}^N$ is the channel from the UE to the RIS, and the diagonal matrix $\mathbf{D}_\psi = \mathrm{diag}(e^{\mathrm{j}\psi_1}, \ldots, e^{\mathrm{j}\psi_N})$ specifies the RIS configuration with $\psi_n \in [-\pi, \pi)$. With maximum-ratio combining at the BS, the SNR with transmit power P and noise power σ^2 is $\mathrm{SNR} = P\|\mathbf{h}\|^2/\sigma^2$. The BS computes the SNR-maximizing RIS configuration and conveys it to the RIS through a feedback message of t bits, transmitted over a control channel with orthogonal time-frequency resources and assumed to be coded so that reception is error-free.

4.4.1 Naive Element-Wise Quantization

The baseline approach is to quantize each phase value in $\mathbf{D}_\psi$ separately using b bits, so that the configuration is represented by $t = Nb$ bits in total. This leads to a feedback load that grows linearly with the number of RIS elements, which is problematic since practical RIS implementations can feature hundreds or even thousands of elements [62]. The codebook developed in this chapter exploits the structure of the cascaded LoS channel to reduce this scaling to logarithmic in N.

4.5 Codebook Design for Cascaded LoS Channels

For pure LoS propagation between the BS and the RIS at angle θ_1, and between the RIS and the UE at angle θ_2, the array gain of the RIS-reflected beam takes the form

$$\mathrm{G_A} = \left| \sum_{n=1}^{N} e^{-\mathrm{j}\pi(n-1)(\sin\theta_1 + \sin\theta_2) + \mathrm{j}\psi_n} \right|^2, \tag{4.2}$$

which is maximized to its peak value N^2 by setting $\psi_n = \pi(n-1)(\sin\theta_1 + \sin\theta_2)$. The crucial observation is that the optimal phase shifts depend on the element index n and the value of

$$\Theta \triangleq \sin\theta_1 + \sin\theta_2 \in (-2, 2). \tag{4.3}$$

It is therefore sufficient for the BS to convey a quantized version $\hat{\Theta}$ of this single scalar; the RIS itself can compute all N phase shifts locally.

If $\hat{\Theta} \approx \Theta$ is received at the RIS and used to set $\psi_n = \pi(n-1)\hat{\Theta}$, the resulting array gain is closely approximated by

$$\mathrm{G_A} \approx N^2 \,\mathrm{sinc}^2\left(\frac{N(\Theta - \hat{\Theta})}{2}\right), \tag{4.4}$$

where $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$. The codebook is constructed by partitioning the interval $(0, 2)$ into $|C| = 2^l$ equal-length bins and using the bin midpoints Θ_j as codewords. The codebook entry $\mathbf{D}_j$ is given by

$$\mathbf{D}_j = \mathrm{diag}\big(1,\, e^{\mathrm{j}\pi\Theta_j},\, \ldots,\, e^{\mathrm{j}\pi(N-1)\Theta_j}\big), \tag{4.5}$$

and the BS transmits the index of the codeword closest to Θ modulo 2.

The half-power beamwidth (HPBW) of the reflected beam is approximately $1.772/N$ [46, Sec. 4.3]. Requiring the worst-case quantization error to keep the receiver inside the HPBW yields the bit budget

$$l = \lceil 0.1746 + \log_2(N) \rceil \quad \text{bits}, \tag{4.6}$$

which is the central scaling result of Paper B. Compared to the Nb-bit element-wise baseline, this is exponentially smaller for large RISs, and the array gain remains within 3 dB of its peak value N^2 regardless of the actual angles θ_1, θ_2. Paper B further shows that the lower bound on the array gain converges to N^2 at rate $O(4^{-x})$ when $l = \log_2(N) - 1 + x$, so each additional bit beyond the HPBW threshold reduces the gap to the array gain by a factor of four.

4.6 Selected Results

4.6.1 Comparison with Element-Wise Quantization Under Rician Fading

In LoS-dominant environments, the channel is more accurately modelled as Rician fading with factor κ rather than pure LoS. Paper B shows that, when the proposed codebook is used, the average channel gain converges to

$$\mathbb{E}\{\|\mathbf{h}\|^2\} = \frac{\kappa}{\kappa+1} KN^2\beta_r\beta_t + \frac{1}{\kappa+1} KN\beta_r\beta_t, \tag{4.7}$$

where β_r and β_t are the RIS-to-BS and UE-to-RIS path gains. The full array gain N^2 is preserved on the LoS component, while the diffuse component contributes only an aperture gain of N. For the typical Third Generation Partnership Program (3GPP) value $\kappa = 10\,\text{dB}$ at a 100 m UE-RIS distance, the LoS term dominates and the codebook incurs only a minor loss.

Fig. 4.2 compares the proposed codebook against element-wise quantization and a DFT-based codebook for $N = 128$ elements. The SNR rises sharply up to the HPBW threshold $l = 8$ and saturates immediately afterwards, in agreement with the convergence analysis. By contrast, achieving comparable SNR with naive element-wise quantization requires $t = 3\times128 = 384$ bits which is a factor of roughly 48 more feedback. As a related result, Paper B shows that grouping every two adjacent RIS elements into a common-phase subarray reduces the average array gain to $\mathbb{E}\{\mathrm{G_A}\} \approx 0.546\,N^2$, corresponding to a 2.63 dB loss before any quantization is applied. This connects directly to the subarray analysis of Chapter 3 and quantifies the price of the lower-overhead grouping scheme used there.

4.6.2 Static Path Synchronization and 3D Geometries

When the static path $\mathbf{h}_s$ is non-negligible, the RIS-reflected signal must be phase-aligned with it to ensure constructive superposition. Paper B introduces an additional common phase rotation ϕ applied to all RIS elements, $\mathbf{D}_{\psi,\phi} = e^{\mathrm{j}\phi}\mathbf{D}_\psi$, and shows that the optimal value $\phi = \arg(\mathbf{a}_K^H(\varphi)\mathbf{h}_s)$ must itself be quantized using d feedback bits. The resulting cross-term in the average channel gain is

$$\mathbb{E}\{2\Re(\mathbf{a}_K^H(\varphi)\mathbf{h}_s e^{-\mathrm{j}\phi})\} = \mathrm{sinc}\left(\frac{1}{2^d}\right) N\sqrt{\pi\beta_r\beta_t K\rho}, \tag{4.8}$$

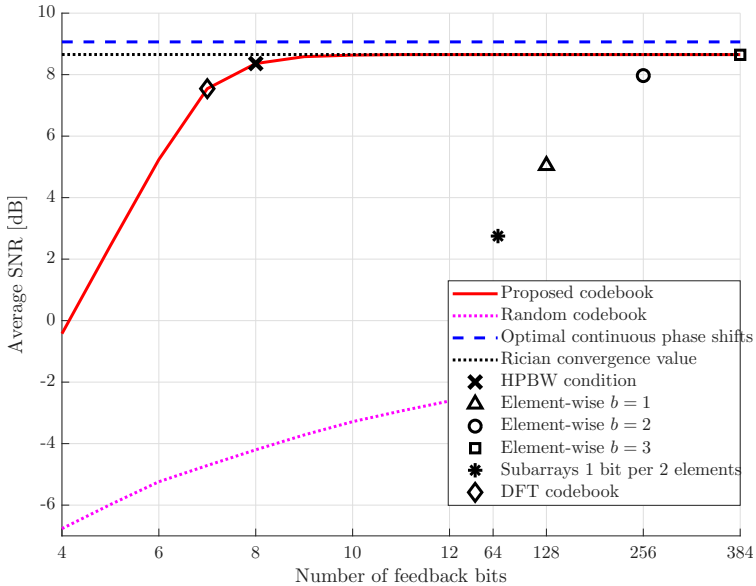


Figure 4.2: Uplink SNR achieved with an RIS of $N = 128$ elements under Rician fading with $\kappa = 10$ dB, using the proposed codebook (solid), a DFT-based codebook, and naive element-wise quantization. The HPBW threshold from (4.6) is reached at $l = 8$ bits. The element-wise baseline requires $t \geq 384$ bits to achieve a comparable SNR.

which collapses to zero for $d = 0$ and approaches its maximum as $d \to \infty$. Paper B further bounds the worst-case SNR loss when d bits are used; for $K = 1$ and an unaligned RIS, the loss reaches up to 2.76 dB, which is significant.

Fig. 4.3 shows the resulting SNR as a function of the total bit budget. Once the HPBW requirement (4.6) is satisfied (here $l \geq 8$), allocating any remaining bits to the static-path alignment ϕ is more valuable than further refining the beam-direction codebook. Two bits for ϕ are essentially sufficient to recover the full SNR.

For a planar UPA-shaped RIS with $N = N_H N_V$ elements in a 3D environment, both azimuth and elevation must be configured. Paper B shows that the relevant scalars $\hat{\Theta}$ and $\hat{\Phi}$ are constrained to lie inside a circle of radius 2 rather than a square (Lemma 5 in Paper B). Exploiting this geometric constraint through a non-uniform circular grid that prioritizes the dimension with more elements yields a gain of up to 5 dB at $l = 8$ bits over independent uniform quantization of the two angles. The total required feedback budget remains

$$t = \log_2(N) + 2 + d \quad \text{bits}, \tag{4.9}$$

which still scales logarithmically with the total number of RIS elements despite the additional angular dimension. For an RIS with $N = 1024$ elements and

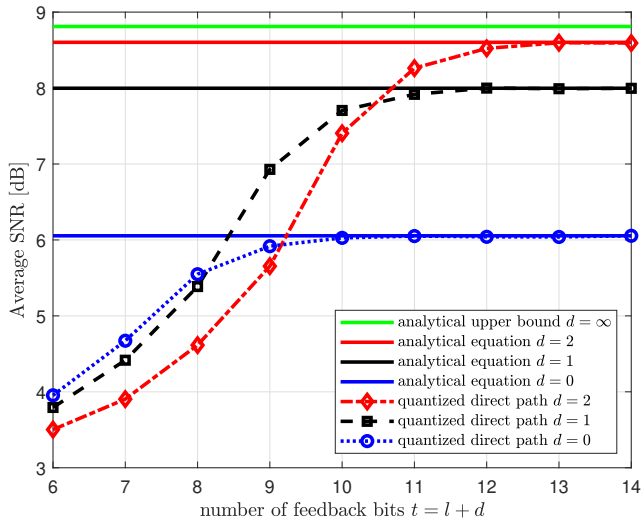


Figure 4.3: Uplink SNR for a single-antenna BS and an RIS with $N = 128$ elements under pure LoS, when $d \in \{0, 1, 2\}$ bits are used to align the RIS reflection with the static path. Allocating $d = 2$ bits is sufficient to virtually close the gap to the unquantized ϕ case once the HPBW requirement on l is met.

$b = 2$, this yields $t = 14$ bits per configuration update, compared to $t = 2048$ bits for naive element-wise quantization, which is a a reduction of more than two orders of magnitude.

4.6.3 UE Mobility, Differential Feedback, and Discrete Phase Shifts

When the RIS serves a single mobile UE over an extended period, the codebook can be updated through lightweight differential commands that exploit temporal channel correlation. Paper B proposes a 2-bit update message `00` keep, `01` shift left, `10` shift right, `11` reconfigure. That suffices in the typical case where the UE moves between neighboring RIS beams. Numerical evaluation over 6000 UEs in a 200×200 m grid moving at 3 m/s shows a clear decoupling of the two update timescales: the static-path alignment ϕ should be refreshed every channel coherence time, while the beam direction $\widehat{\Theta}$ can be updated much less frequently, on the order of the beam coherence time. Frequent ϕ updates produce the largest SNR gains because misalignment with the static path can otherwise cause destructive superposition.

Practical low-cost RIS hardware uses PIN-diode-based elements [63] that can only realize 2^b discrete phase shifts per element. Paper B derives the combined effect of codebook quantization and discrete-phase hardware on the expected

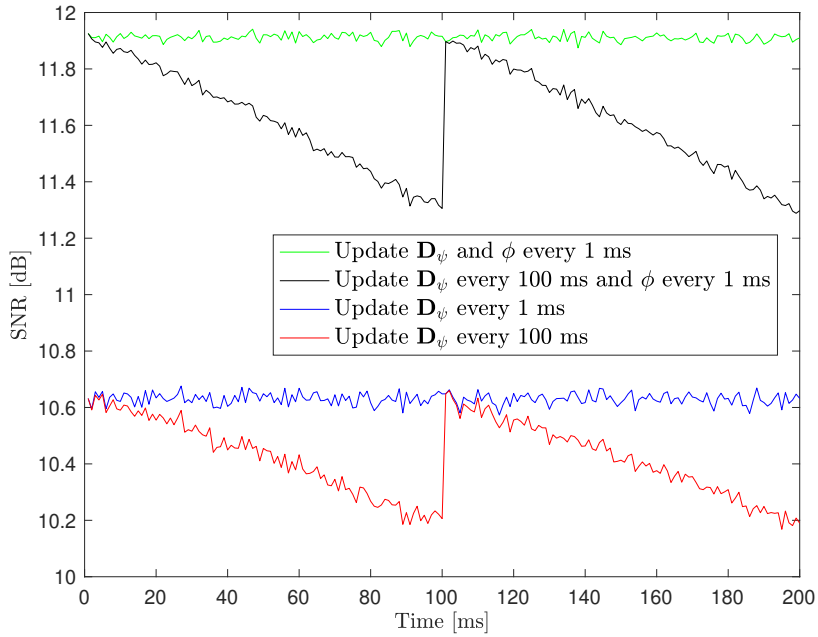


Figure 4.4: Achieved SNR with the proposed differential update approach. The RIS configuration matrix component that aligns the RIS path with the static path and the RIS configuration matrix part responsible for the general beam direction are updated at different timings.

array gain as

$$\mathbb{E}\{\mathrm{G_{AD}}\} \approx \mathbb{E}\{\mathrm{G_A}\}\,\mathrm{sinc}^2\left(\frac{1}{2^b}\right) + \sqrt{\mathbb{E}\{\mathrm{G_A}\}}\left(1 - \mathrm{sinc}^2\left(\frac{1}{2^b}\right)\right), \tag{4.10}$$

where $\mathbb{E}\{\mathrm{G_A}\}$ is the continuous-phase array gain. Fig. 4.5 shows that the analytical approximation matches Monte Carlo simulations closely, and that the HPBW design rule $l = \lceil \log_2(N) + 0.17 \rceil$ remains a good operating point under discrete-phase hardware. Setting $d = b$ i.e., matching the static-path alignment resolution to the hardware phase resolution, is recommended as a practical design choice, since it minimizes the RIS-vs-static-path misalignment without exceeding what the hardware can actually realize.

4.7 Summary

The implementation of RIS technology in real networks requires not only good channel estimation but also resource-efficient control signaling to convey the resulting configuration to the RIS. By tailoring the codebook to the geometric structure of the cascaded LoS channel, the feedback load can be reduced

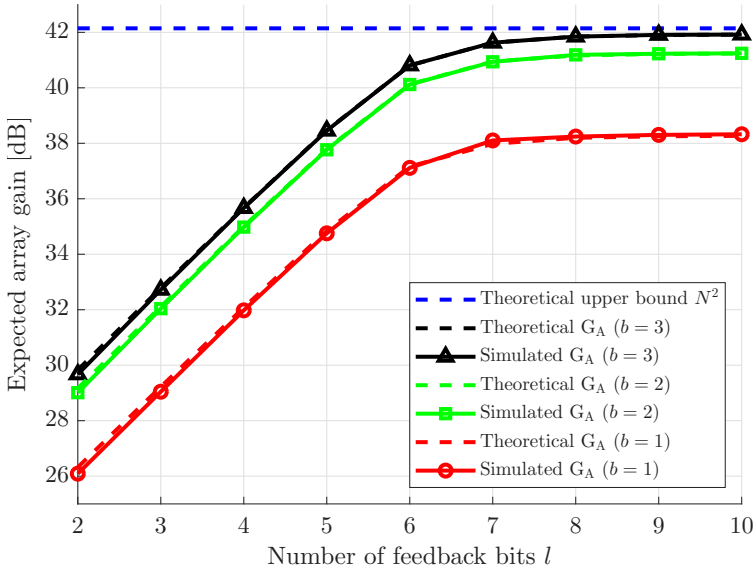


Figure 4.5: Expected array gain for an RIS with $N = 128$ elements with continuous phase shifts and with discrete phase shifts of $b \in \{1, 2, 3\}$ bits per element. The HPBW threshold $l = 8$ remains a good operating point even under discrete-phase hardware. The analytical approximation (4.10) matches Monte Carlo simulations closely.

from linear to logarithmic scaling with the number of RIS elements, while keeping the SNR loss within the HPBW. The same design principle extends to 3D UPA geometries through a non-uniform circular quantization grid, to non-zero static paths through a separate phase-alignment parameter, and to mobility through a 2-bit differential update with decoupled timescales. The combined SNR-loss analysis quantifies how the codebook resolution interacts with the discrete phase-shift hardware of practical RIS implementations, and identifies $d = b$ as a sound design choice. The overall takeaway is that an RIS designed to create a virtual LoS path between BS and UE, which is the deployment scenario in which RIS is most useful in the first place [64], requires only a logarithmic amount of feedback signaling, making the technology suitable for deployment with limited control-channel resources.

5 Power Consumption Modeling and Energy Efficiency Limits

This chapter develops the PC framework utilized in the optimization-oriented chapters of the thesis. We first introduce the hardware components of a radio transmitter and translate their behavior into three nested PC models of increasing generality: a single-carrier single-antenna model, a single-carrier multi-antenna model, and a multi-carrier extension. We then layer sleep modes on top of these models to capture the average PC of a transceiver that is not always actively transmitting. Finally, we use the simplest of the PC models to derive a fundamental upper bound on the EE achievable by any wireless link, which serves as a reference point for the more refined optimizations carried out in Chapters 6, 8, and 7.

5.1 Power Consumption Modeling

Although all parts consume power in wireless networks, the most significant part is the radio access network (RAN) [21]. In the RAN, the most energy-consuming part is the radio transmitter, which is composed of several components needed to process and transmit a signal. The PC model used in this thesis is built up from these individual components and is adapted to the multi-carrier model in [65], itself a refinement of the earlier component-level model in [66]. A complementary machine-learning model fitted to real measurement data [23] confirmed that a linear analytical model of the form used here is a sufficiently accurate representation of the actual PC.

We build the model in three stages of increasing generality. Section 5.1.1 introduces the hardware components and their PC contributions. Sections 5.1.2–5.1.4 then assemble these contributions into three nested models: a single-carrier

single-input single-output (SISO) model used for the EE upper-bound analysis in Section 5.3, a single-carrier multi-antenna model used in Chapters 6 and 7, and a multi-carrier extension used in Chapter 8.

5.1.1 Hardware Components and Their Contributions

Here we briefly state the function of the different hardware components and how their contribution to the total PC is modeled. We focus on the radio transmitter. Fig. 5.1 presents a detailed view of its components. The first stage is the digital signal processing (DSP) module, tasked with handling the digital baseband (BB) processing of the physical layer (modulation, coding, resource mapping) along with medium-access functions such as re-transmission control. Following this, the BB signal is converted to analog form via a digital-to-analog converter (DAC), passed through a low-pass filter (LPF) to eliminate high-frequency components introduced by the DAC [67], up-converted from intermediate frequency (IF) to RF by a mixer driven by a local oscillator (LO), filtered by a band-pass filter (BPF), and finally amplified by the PA before being radiated by the antenna.

All components that consume energy, including signal processing and analog and digital circuitry, play a role in determining the optimal operating point. This point varies across BS models with different characteristics and shifts over time as hardware properties improve. Nevertheless, a basic physical constraint, the Landauer limit, sets a minimum on the EC required for switching, which is important for computational power [68, 69].

- **Power amplifier (PA).** The PA amplifies the low-power RF signal so it can be transmitted over long distances. Its efficiency is the ratio of output power (radiated to the antenna) to input power, and we characterize the PA by its efficiency factor

$$\kappa = \frac{\text{output power}}{\text{input power}}. \tag{5.1}$$

  Designing efficient PAs involves trade-offs with linearity, bandwidth, complexity, and cost. We do not separately model feeding losses between the PA and the antenna, since they are small in modern integrated radios; any residual losses can be absorbed into κ. A small constant circuit power required to keep the PA on is folded into the constant μ introduced below.

- **Local oscillator (LO).** The LO generates a stable carrier-frequency reference used for up- and down-conversion. Its PC is well modelled as a constant contribution to the total fixed PC μ. Different LOs are typically used in different frequency ranges, so a BS aggregating a sub-6 GHz carrier with a higher-band carrier requires two separate LOs, each with its own (typically increasing-with-frequency) PC. This carrier-specificity

has implications for multi-carrier deployments and is one reason multi-carrier PC must be modelled per band; we return to this in Section 5.1.4.

- **Digital signal processor (DSP).** The DSP runs the BB algorithms. It has a fixed PC component (folded into μ) and a component proportional to the sampling rate, which in turn is proportional to the bandwidth B. The DSP therefore contributes an additional νB to the total PC, for some hardware constant $\nu > 0$.
- **Digital-to-analog and analog-to-digital converters (DAC/ADC).** The converters operate sample-by-sample, so their PC is also bandwidth-proportional and is absorbed together with the DSP contribution into νB. Their constant standby PC is similarly absorbed into μ.

The numerical values of κ, μ, and ν depend on the specific hardware and on the carrier frequency, are not generally publicly disclosed by vendors, and improve over time as manufacturing processes mature. We treat them as fixed but generic constants throughout the analytical work in this thesis.

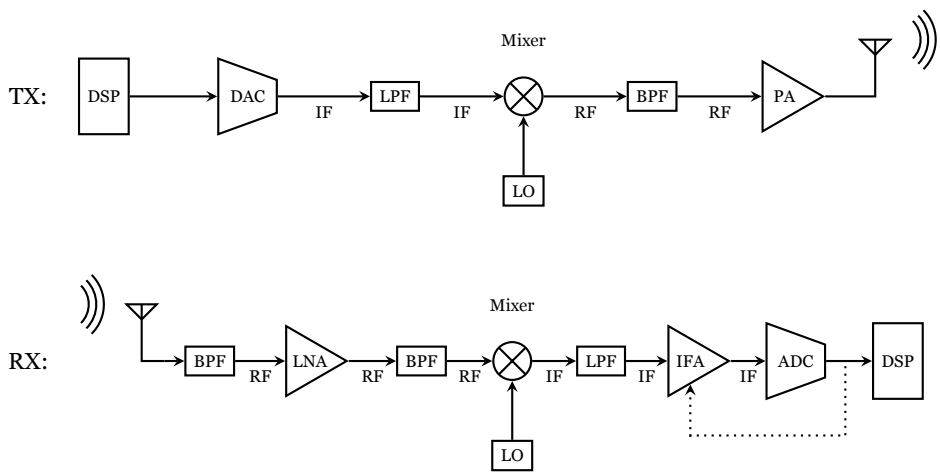


Figure 5.1: Radio unit block diagram. Reproduced with permission from [70].

5.1.2 Single-Carrier, Single-Antenna Model

A simple setup considers a single-antenna transmitter on a single carrier. The total PC is

$$P_{\text{total}} = \frac{P}{\kappa} + \mu + \nu B, \tag{5.2}$$

where P is the radiated transmit power and the three terms represent the active radiated/dissipated power, the fixed circuit and synchronization PC, and the

bandwidth-proportional sampling and processing PC, respectively. SISO links remain practically relevant for low-band carriers below 1 GHz, where coverage is prioritized over peak rate, and for control signaling in higher-band systems where simplicity and robustness matter more than aggregate throughput. We use this model in Section 5.3 to establish the fundamental EE upper bound that motivates the rest of the thesis.

5.1.3 Single-Carrier, Multi-Antenna Model

When the BS is equipped with M transmit antennas and the rate-dependent BB processing cost cannot be neglected, (5.2) extends to

$$P_{\text{total}} = \frac{P}{\kappa} + \mu + (D_0 + \nu B)M + \eta R, \tag{5.3}$$

where $D_0 \geq 0$ is the per-antenna passive PC of one transceiver chain, $\eta \geq 0$ scales the rate-dependent encoding/decoding PC at the BS, and R is the achieved data rate. The per-antenna term $D_0 M$ captures the cost of replicating the analog front-end across antenna ports, while νBM captures the proportional growth of the per-chain DAC/ADC and BB processing cost with bandwidth. When the rate is set to the Shannon capacity over a multiple-input single-output (MISO) link with array gain, $R = B\log_2(1 + MP\beta/(BN_0))$, (5.3) becomes the model used throughout Chapters 6 and 7. Setting $M = 1$ (absorbing D_0 as a fixed constant into μ) and $\eta = 0$ recovers (5.2).

5.1.4 Multi-Carrier Extension

When a BS operates over multiple carriers simultaneously, (5.3) extends by treating each carrier as its own transceiver chain with its own constants. For a site with K carriers indexed by $i \in \{1, \dots, K\}$, the total PC is

$$P_{\text{total}} = \sum_{i=1}^{K} \left[\frac{P_i}{\kappa_i} + \mu_i + (D_{0,i} + \nu_i B_i)M_i + \eta_i R_i \right], \tag{5.4}$$

where each band i carries its own PA efficiency κ_i, fixed cost μ_i, per-antenna passive cost $D_{0,i}$, bandwidth-proportional cost ν_i, rate-dependent cost η_i, transmit power P_i, bandwidth B_i, and antenna count M_i.

The per-carrier decoupling in (5.4) is appropriate for inter-band carrier aggregation (CA) and for non-contiguous intra-band CA, where the PA, LO, data converters, and BB DSP are replicated per carrier because their operating frequency and sampling rate are carrier-specific, and where κ_i, $D_{0,i}$, and $\nu_i B_i$ all depend strongly on the carrier frequency and are typically poorer at higher bands. For contiguous intra-band CA, by contrast, a single wideband front-end can cover both component carriers and several of these costs are shared rather

than duplicated; (5.4) can still be used in that case, but with the caveat that the per-band constants are no longer fully independent.

The multi-carrier framework was introduced in this form by [65], who provide measurement-aligned values for the per-band constants across sub-6 GHz and higher-band massive MIMO transceivers. We adopt their per-carrier decoupling in Chapter 8, which considers the inter-band case $K = 2$ with a sub-6 GHz primary carrier and a sub-THz secondary carrier.

5.2 Sleep Modes and Average Power Consumption

The PC models in Sections 5.1.2–5.1.4 describe the instantaneous PC of an actively transmitting transceiver. In practice, traffic is bursty (see Chapter 7), and a BS that is not actively transmitting can power down parts of its hardware to conserve energy. We model this by introducing an *activity factor* $\alpha \in [0, 1]$, defined as the fraction of time the transceiver is actively transmitting, and a *sleep power* P_s, which is the PC during the remaining $(1 - \alpha)$ fraction of time.

The average PC over a sufficiently long observation window is then

$$\bar{P} = \alpha P_a + (1 - \alpha) P_s, \tag{5.5}$$

where P_a is the active PC given by whichever of (5.2), (5.3), or (5.4) applies. The sleep power depends on the depth of the sleep state and is parametrized as $P_s = \delta_s(\mu + D_0 M)$ for a single-carrier multi-antenna transceiver, where $\delta_s \in [0, 1]$ controls the fraction of passive PC that is retained while sleeping [71]. Three canonical sleep modes are commonly distinguished, ordered by increasing depth and increasing wake-up latency τ:

- **Idle mode** ($\delta_s = 1$, $\tau \approx 0$). Circuitry and transceiver chains are powered, ready to transmit on the next slot. No power is saved relative to active idle, but no wake-up delay is incurred.
- **Deep sleep** ($\delta_s = \mu/(\mu + D_0 M)$, $\tau \sim$ ms). Per-antenna chains are powered down, only the fixed circuitry μ remains.
- **Absolute sleep** ($\delta_s = 0$, $\tau \sim$ tens of ms). All hardware is powered down. The BS consumes no PC while sleeping but wake-up time is the longest.

The trade-off between sleep depth (energy saved) and wake-up latency (QoS impact) is analyzed in Chapter 7. The framework extends to the multi-carrier model in (5.4) by attaching a per-band activity factor α_i and a per-band sleep power $P_{s,i}$, allowing each carrier to enter and exit sleep independently of the others. This per-band sleep coordination and dual-band activation threshold is analyzed further in Chapter 8.

5.3 Upper Limits of Energy Efficiency

In this section we use the single-carrier single-antenna model of (5.2) to establish a fundamental upper bound on the EE achievable by any wireless link, which serves as a reference point for the more refined optimizations carried out in later chapters.

In his groundbreaking paper [72], Claude Shannon formulated the capacity of mathematical communication in the presence of noise, which defines how many bits per second can be transmitted in a given bandwidth. There are key assumptions made, namely that the block length of the data is infinite and a channel code is employed with a decoding error rate that approaches zero. The capacity is defined as

$$C = B \log_2(1 + \text{SNR}). \tag{5.6}$$

Where we have SNR being defined as the received signal power divided by the thermal noise in the receiver. The received power of a signal transmitted with power P over a radio channel with gain β is $P\beta$. The thermal noise power is defined as BN_0, where B is the bandwidth over which the receiver listens to the signal. $N_0 = k_B T$ is the thermal noise power spectral density defined by Boltzmann's constant $k_B = 1.380649 \cdot 10^{-23}\,\text{m}^2\,\text{kg}\,\text{s}^{-2}\,\text{K}^{-1}$ and T is the receiver temperature in Kelvin, which is assumed to be at room temperature of 293.15 K throughout this thesis.

The EE is defined as the number of bits transmitted per unit of energy [bit/J]:

$$\text{EE} = \frac{R}{P} \tag{5.7}$$

where P denotes the power consumed and R is the achieved data rate, which is upper bounded by C. Generally, the gap between R and C is larger for smaller payloads [73].

The discussions in [74, 75] address the physical and practical constraints on the EE of wireless communication systems, highlighting that operating in a wideband regime with low SNR is most favorable. It is interesting to note what happens as one looks at the EE metric, especially how it varies depending on the detail of the PC model.

5.3.1 Transmit Power Model

The most simple model of the EE in a SISO channel only considers the transmitted power. SISO channels continue to carry a great deal of importance for systems operating in the scarce low band frequencies below 1 GHz where fewer antennas are utilized. Low frequencies provide great coverage albeit at the expense of less available bandwidth. Many existing communication systems and standards still rely on SISO architectures and even in advanced communication systems, such as 5G, that primarily use MIMO for data transmission,

SISO channels are often used for control signaling due to their robustness and simplicity. The SISO example has already been considered in the literature [75].

If only the transmitted power P is considered, the EE becomes:

$$\mathrm{EE} = \frac{B \log_2(1 + \mathrm{SNR})}{P}. \tag{5.8}$$

The deterministic SNR for the SISO-channel is $\mathrm{SNR} = \frac{P\beta}{BN_0}$, where P is the transmitted power, B is the bandwidth, β is the channel gain, and N_0 is the noise power spectral density. We note that the EE is maximized in the limit as $P/B \to 0$. This gives the maximum EE as

$$\mathrm{EE}_{\max} = \frac{\beta}{N_0 \ln(2)}. \tag{5.9}$$

The achieved rate goes to zero as well, meaning that the most energy-efficient system works impossibly slowly for any real-world applications. The consequence is that to maximize the EE, the transmission should be *very* slow. However, this scenario illustrates that a more detailed hardware model is required for the results to be reasonable.

5.3.2 Circuit Power Inclusion

A more realistic model over the total PC P_{total} is

$$P_{\text{total}} = P + \mu + \nu B \tag{5.10}$$

where $\mu > 0$ is the circuit power needed to run the analog and digital circuits of the transceiver and $\nu B > 0$ is the power consumption of the ADC and DAC converters of the samples plus the BB processing of the samples. This means that the required power for converting and processing the samples is proportional to some constant $\nu > 0$. In this case, there exists an upper bound of the EE for a certain ratio of P/B such that:

$$\frac{P}{B} = N_0 \frac{e^u - 1}{\beta}, \tag{5.11}$$

where

$$u = W\left(\frac{\beta\nu}{N_0 e} - \frac{1}{e}\right) + 1 \tag{5.12}$$

and the resulting upper bound is

$$\mathrm{EE}_{\max} \leq \frac{u \log_2(e)}{N_0 \frac{e^u - 1}{\beta} + \nu} \tag{5.13}$$

$W(\cdot)$ denotes the Lambert W function, defined by the equation $x = W(x)e^{W(x)}$ for any $x \in \mathbb{C}$ [76].

EE_{max} is achieved as both $P \to \infty$ and $B \to \infty$ jointly while fulfilling this ratio. It is interesting to note that the factor μ plays no important role in this limit since it is made negligible assuming there is enough transmit power and bandwidth available.

5.4 Summary

This chapter has assembled the analytical machinery used in the rest of the thesis. The three nested PC models in (5.2)–(5.4) progressively add antenna arrays and multiple carriers while preserving the same parametric form, which makes it possible to treat the SISO upper-bound analysis of Section 5.3, the joint (P, B, M) optimization in Chapter 6, the dual-band coordination in Chapter 8, and the latency-constrained sleep-mode scheduling in Chapter 7 within a single framework. The sleep-mode parametrization $P_s = \delta_s(\mu + D_0 M)$ in turn unifies idle mode, deep sleep, and absolute sleep under a single activity-factor expression (5.5).

6 Scaling Behaviors and Optimal Ratios for Energy Efficient Downlink Communication

Our second paper presents an in-depth analysis of the fundamentals of optimizing EE in wireless communication links, particularly focusing on BSs equipped with multiple antennas. We utilize the sophisticated PC models we discussed in Chapter 2 earlier, incorporating both passive and active components of the transceiver circuitry. This includes considerations for radiated power, signal processing, and passive circuit power consumption. We also seek to identify tangible relationships between these parameters at the optimal solution. Furthermore, this thesis focuses on the scaling behavior of EE and its solution, meaning how the EE behaves as the power or bandwidth can be allowed to be arbitrarily large. We conduct an in-depth analytical analysis of hardware characterizing constants and their impact on EE, shedding new light on current state-of-the-art EE models.

There are inherent physical limitations. The ideal case is, of course, a lossless transmission from the transmitter to the receiver which captures all the transmitted power. Such an implementation could be made by a theoretical lossless cable. Alternatively, a receiving antenna that encloses the transmitting antenna and thus captures all the received signal energy would also work. However, such examples offer no help in understanding what the limits on EE may be in practical scenarios. In this thesis, we have therefore focused on including the PC of components that can be considered essential in modern wireless systems that one could not imagine getting rid of in the foreseeable future.

6.1 Research Question

The main research questions are listed below.

- How should the transmit power, bandwidth, and number of antennas be jointly configured to maximize EE?
- Are there any tangible relationships between these parameters at the optimal solution?

6.2 Related Works

There is a considerable effort being spent on reducing the EC in the networks of MNOs today. This has involved standardization efforts in 3GPP when evolving the network from LTE to New Radio (NR). The main gains so far have improved and implemented new techniques of sleep mode, lean carrier, selective hardware shutdown etc. as detailed in Chapter 2. As a result, many previous studies have focused on practical network EC needs and improvement.

Two papers that employ analytical analysis are [66], which deals with PC models for massive MIMO and how to optimize them, and [75], which gives some upper limits on how EE a wireless communication system can become and also discusses power per bandwidth, albeit under a more simple PC model. Some further papers that examine the gains of EE that can be made have examined particular pieces of hardware and the match of analytical models. Alternatively, the area PC under rate constraints can be optimized through advanced algorithms [77]. In summary, the research area of fundamental EE limits and optimal operating points is relatively unexplored, which our work aims to shed new light on.

6.3 Contributions and Methodology

We analyze the EE over a single wireless link, with a focus on how far we can push this metric given a set of hardware characterizing constants. We study the impact of these on the solution and consider all sources of EC.

We mathematically formulate a non-linear optimization problem of the EE in a multi-antenna BS utilizing a single carrier. Through classical analysis, we derive results on optimal ratios of power per unit bandwidth and find a relationship between the transmitted power and passive transceiver PC in a BS. Furthermore, we use our analytical and numerical results to create a robust algorithm that quickly converges to find the global optimum, thus solving the optimization problem under constant constraints on the variables.

6.4 System Model

We consider a MISO channel, with one component carrier with bandwidth B. The rate achieved on the MISO-channel is given by the capacity upper bound, with each path having a magnitude gain of β over the carrier.

6.4.1 Power Consumption Model

We adopt the single-carrier multi-antenna PC model of Section 5.1.3, restated for convenience as

$$\text{PC} = \frac{P}{\kappa} + \mu + (D_0 + \nu B)M + \eta R, \tag{6.1}$$

with $R = B\log_2(1 + MP\beta/(BN_0))$ the rate achieved on the MISO link. The corresponding EE expression is

$$\text{EE} = \frac{B\log_2\left(1 + \frac{MP\beta}{BN_0}\right)}{P/\kappa + \mu + (D_0 + \nu B)M + \eta B\log_2\left(1 + \frac{MP\beta}{BN_0}\right)}, \tag{6.2}$$

which increases monotonically with the bandwidth B.

6.4.2 Optimization Problem

The goal is to solve the following joint EE maximization problem:

$$\begin{aligned} \underset{P,B,M}{\text{maximize}} \quad & \text{EE} \\ \text{subject to} \quad & 0 < P \leq P_{\max}, \\ & 0 < B \leq B_{\max}, \\ & M \in \{1, \dots, M_{\max}\}, \end{aligned} \tag{6.3}$$

with the EE defined as in (6.2). The variables transmit power P, bandwidth B, and number of antennas M are constrained by upper bounds $(P_{\max}, B_{\max}, M_{\max})$.

We prove that (6.3) is solved at the boundary where $P = P_{\max}$ or $B = B_{\max}$. This means that more bandwidth does not necessarily increase the EE unless one can also simultaneously transmit using more total power.

The EE objective in (6.2) is the ratio of two non-linear functions of (P, B, M) and is not jointly separable in the three variables since the SNR term $MP\beta/(BN_0)$ couples power, antennas, and bandwidth inside the logarithm, and the bandwidth B enters the PC denominator both through the per-antenna term νBM and through the rate-dependent processing term $\eta B\log_2(1 + MP\beta/(BN_0))$. It has therefore no closed-form solution. The integer constraint $M \in \{1, \dots, M_{\max}\}$ further excludes purely continuous methods.

As we show in Sections 6.5 and 6.7, the EE is a unimodal function of the remaining variable in each case, and its maximizer can be expressed via the principal branch W_0 of the Lambert W function. These per-variable optimizers, combined with the boundary observation following Theorem 1 that at least one of $P = P_{\max}$ or $B = B_{\max}$ must bind at the global optimum, are the building blocks of an alternating optimization procedure that converges rapidly to the joint optimum. The integer variable M can be handled by including it in the alternating updates via the expression in (6.17) and by quantizing it as needed.

6.5 Optimal Power-to-Antenna Ratio

It is possible to analytically optimize the transmit power per antenna P/M for a given value of B. We show how such a ratio exists. The argument given here is an alternative proof of the AM-GM-based result used in Paper C.

Theorem 1. *If the solution $(P_{\text{opt}}, M_{\text{opt}})$ that maximizes the EE in (6.2) for a given value of B satisfies $P_{\text{opt}} \leq P_{\max}$ and $M_{\text{opt}} \leq M_{\max}$, then the following relation holds:*

$$\frac{P_{\text{opt}}}{M_{\text{opt}}} = \kappa(D_0 + \nu B). \tag{6.4}$$

Equation (6.4) is equivalent to the realization that the transmitted power plus heat dissipated in the PA equals the passive circuitry PC, by a simple rearrangement of the terms:

$$\frac{P_{\text{opt}}}{\kappa} = (D_0 + \nu B)M_{\text{opt}}. \tag{6.5}$$

That is, at the EE-optimum the active power consumption (radiated power plus PA dissipation) is matched exactly by the passive transceiver consumption.

Proof. The transmit power P and the number of antennas M enter the capacity in (6.2) only through their product MP. For any fixed B, the achievable rate, the rate-dependent processing cost $\eta B \log_2(1 + MP\beta/(BN_0))$, and the fixed term μ are therefore all functions of MP alone. Minimizing the total power consumption for a given target rate thus reduces to minimizing the only remaining P, M-dependent contribution, namely

$$f(P, M) = \frac{P}{\kappa} + (D_0 + \nu B)M, \tag{6.6}$$

subject to $MP = K$ for some constant $K > 0$ determined by the rate target. Substituting $M = K/P$ yields the single-variable function

$$f(P) = \frac{P}{\kappa} + \frac{K(D_0 + \nu B)}{P}, \tag{6.7}$$

which is defined on $P > 0$.

Differentiating and setting the derivative to zero gives

$$\frac{\mathrm{d}f}{\mathrm{d}P} = \frac{1}{\kappa} - \frac{K(D_0 + \nu B)}{P^2} = 0, \tag{6.8}$$

so the unique stationary point on $P > 0$ satisfies $P^2 = K\kappa(D_0 + \nu B)$. The second derivative is

$$\frac{\mathrm{d}^2 f}{\mathrm{d}P^2} = \frac{2K(D_0 + \nu B)}{P^3} > 0 \tag{6.9}$$

for all $P > 0$, since K, D_0, ν, B are non-negative and $D_0 + \nu B > 0$. The function $f(P)$ is therefore strictly convex on $(0, \infty)$, and the stationary point is the unique global minimum. Substituting $K = MP$ back yields

$$\frac{1}{\kappa} = \frac{MP(D_0 + \nu B)}{P^2} = \frac{M(D_0 + \nu B)}{P}, \tag{6.10}$$

and rearranging gives the claimed relation

$$\frac{P}{M} = \kappa(D_0 + \nu B). \tag{6.11}$$

□

6.6 Optimal Power-to-Bandwidth Ratio

We now show that, in addition to the power-to-antenna ratio of Theorem 1, there is also an optimal ratio of P/B for a given number of antennas M. With the EE growing, larger values of P/B fulfill this ratio, making the constant circuit power consumption terms negligible. However, in practice, there are fixed constraints on maximum power and available bandwidth. Under such fixed constraints, our algorithm converges to the global optimal solution. At the optimal point, a consequence of the optimal ratio under finite constraints is that either power or bandwidth should be maximized.

The optimal ratio of P/B which maximizes the EE in (6.2) is

$$\frac{P}{B} = N_0 \frac{e^u - 1}{M\beta}, \tag{6.12}$$

where

$$u = W_0\left(\frac{\kappa M^2 \beta \nu}{N_0 e} - \frac{1}{e}\right) + 1 \tag{6.13}$$

and $W_0(\cdot)$ denotes the principal branch of the Lambert W function. The relation is illustrated in Fig. 6.1.

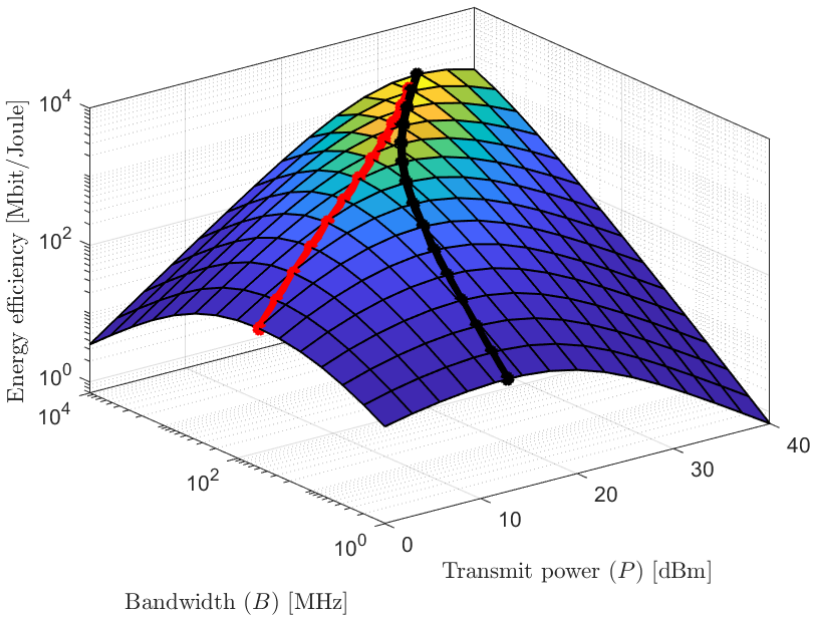


Figure 6.1: The EE is shown as a function of P and B. In black, we have the optimal P for a given B as in (6.14). In red, we have the optimal B for a given P as in (6.16). P and B converge to the optimal ratio P/B given in (6.12). We consider 20 transmit antennas.

6.7 Optimization Algorithm

To create the algorithm, the first step is to optimize the EE with respect to each of the variables P, M, and B when the other ones are fixed.

The first result is: the EE in (6.2) for a given B, M is maximized with respect to P by

$$P = BN_0 \frac{e^v - 1}{M\beta}, \tag{6.14}$$

where

$$v = W_0\left(\frac{\kappa M\beta(\mu + (D_0 + \nu B)M)}{BN_0 e} - \frac{1}{e}\right) + 1. \tag{6.15}$$

The second result is: the EE in (6.2) is a unimodal function of B (for any fixed P, M) that is maximized at a unique B by numerically solving the equation

$$\begin{aligned}&\left(\frac{BN_0}{MP\beta}(\kappa\mu + D_0 M\kappa + P) + \kappa\mu + D_0 M\kappa + P\right)\\ &\times \log_e\left(1 + \frac{MP\beta}{BN_0}\right) = M\kappa\nu B + \kappa\mu + D_0 M\kappa + P.\end{aligned} \tag{6.16}$$

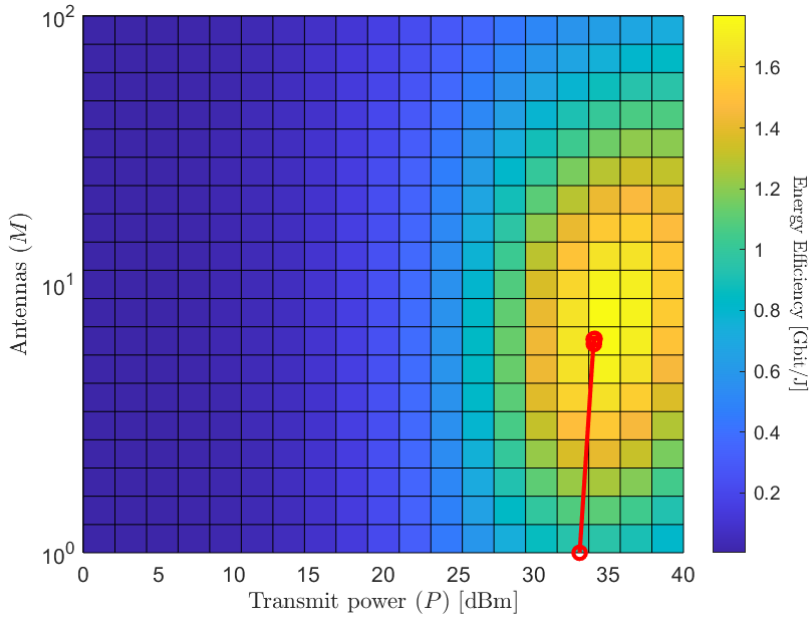


Figure 6.2: The EE is shown as a function of P and M. It converges to the optimum point as detailed in Algorithm 1. Encircled in red are the algorithm's updates of P and M and the corresponding EE.

These two results are shown in Fig. 6.1.

The third result is: the EE in (6.2) for a given P, B is maximized with respect to M by

$$M = BN_0 \frac{e^w - 1}{P\beta}, \tag{6.17}$$

where

$$w = W_0\left(\frac{P\beta(P/\kappa + \mu)}{BN_0 e(D_0 + \nu B)} - \frac{1}{e}\right) + 1. \tag{6.18}$$

These three results, coupled together with the boundary observation following Theorem 1 that at the optimum point $P = P_{\max}$ or $B = B_{\max}$, are used as building blocks to design an alternating optimization algorithm for solving the problem (6.3). The algorithmic iterations are shown in Fig. 6.2 for the changes in the value of transmit power and number of antennas.

6.8 Equivalence of High-Rate Solution to the Global Optimum

We make some remarks here on the similarity of optimizing the unconstrained EE problem to optimizing the EE in the so-called high-rate region and their

equivalence at the optimum rate. This part connects the analysis of the regions and shows their equivalence. It was left out of Paper C for brevity.

Proposition. *For a system where the unconstrained global optimum found by Algorithm 1 lies on the bandwidth boundary ($B_{\text{opt}} = B_{\text{max}}$), evaluating the high-rate closed-form equations at the optimal rate $R = R_{\text{opt}}$ yields exactly the optimal hardware configuration: $P_{\text{high}}(R_{\text{opt}}) = P_{\text{opt}}$ and $M_{\text{high}}(R_{\text{opt}}) = M_{\text{opt}}$.*

Proof. Let $(P_{\text{opt}}, B_{\text{opt}}, M_{\text{opt}})$ be the optimal configuration returned by Algorithm 1 that achieves the maximum energy efficiency at the optimal rate R_{opt}. Assuming the optimal bandwidth is $B_{\text{opt}} = B_{\text{max}}$, the rate is defined by the capacity equation:

$$R_{\text{opt}} = B_{\text{max}} \log_2 \left(1 + \frac{M_{\text{opt}} P_{\text{opt}} \beta}{B_{\text{max}} N_0} \right). \tag{6.19}$$

Because $(P_{\text{opt}}, M_{\text{opt}})$ is the unconstrained globally optimal configuration, it must satisfy the condition established in Theorem 1. Therefore, the optimal power-to-antenna ratio is:

$$\frac{P_{\text{opt}}}{M_{\text{opt}}} = \kappa(D_0 + \nu B_{\text{max}}) \implies M_{\text{opt}} = \frac{P_{\text{opt}}}{\kappa(D_0 + \nu B_{\text{max}})}. \tag{6.20}$$

Substituting this expression for M_{opt} into the capacity equation gives:

$$R_{\text{opt}} = B_{\text{max}} \log_2 \left(1 + \frac{P_{\text{opt}}^2 \beta}{\kappa(D_0 + \nu B_{\text{max}}) B_{\text{max}} N_0} \right). \tag{6.21}$$

Defining the constant c:

$$c = \frac{\beta}{\kappa(D_0 + \nu B_{\text{max}}) B_{\text{max}} N_0}. \tag{6.22}$$

Substituting c into the rate equation yields:

$$R_{\text{opt}} = B_{\text{max}} \log_2(1 + c P_{\text{opt}}^2). \tag{6.23}$$

Inverting this equation to solve algebraically for P_{opt} gives:

$$2^{R_{\text{opt}}/B_{\text{max}}} = 1 + c P_{\text{opt}}^2 \tag{6.24}$$

$$P_{\text{opt}} = \sqrt{c^{-1} \left(2^{R_{\text{opt}}/B_{\text{max}}} - 1 \right)}. \tag{6.25}$$

This derived expression for P_{opt} is exactly the function P_{high} evaluated at $R = R_{\text{opt}}$.

Furthermore, since the high-rate antenna count is defined as $M_{\text{high}} = P_{\text{high}}/[\kappa(D_0 + \nu B_{\max})]$, and we have shown $P_{\text{high}}(R_{\text{opt}}) = P_{\text{opt}}$, it follows directly from Theorem 1 that $M_{\text{high}}(R_{\text{opt}}) = M_{\text{opt}}$. This proves that the high-rate equations at R_{opt} and the Algorithm 1 optimum coincide. □

6.9 Summary

A novel relationship between radiated power and passive transceiver power consumption is uncovered. We provide insights into the optimality of using maximum power or bandwidth and how many transmit antennas are optimal for maximum EE.

We find that when considering a single-layer transmission, at this optimum, maximizing either power or bandwidth is ideal, but the number of transmission antennas should remain finite.

7 Sleep Mode Management and Dynamic Scheduling for Bursty Traffic

7.1 Introduction

In Chapter 6, we established the fundamental EE limits of a fully active multi-antenna BS, deriving the optimal continuous configuration $(P_{\text{opt}}, B_{\text{opt}}, M_{\text{opt}})$ and the corresponding active transmission rate R_{opt}.

In practice, the strategy for managing PC at a BS is also dictated by the traffic characteristics and associated QoS constraints. Modern downlink traffic is dominated by small, bursty sessions: 95% of sessions carry less than 1 Mbit, while only 1% of sessions account for over 70% of the total data volume [8]. Maximizing EE in a realistic deployment requires the BS to dynamically adapt a joint transmission-and-sleep strategy, balancing the energy savings of deeper hardware sleep against the latency penalty of waking up.

This chapter extends the static EE optimization of Chapter 6 into a traffic-aware framework. We integrate three depths of hardware sleep into the PC model, characterize how QoS rate constraints reshape the EE-optimal hardware configuration, and quantify the impact of wake-up transition delays on the viable sleep depth under strict latency budgets. These parts summarize Paper C, whose scope ends with the energy-optimal configuration of a single transmission instance. The final section applies that result to bursty traffic, and shows that the coupling between sleep depth and transmit rate carries over to the scheduling problem.

7.2 Research Questions

Building on the continuous hardware optimization of Chapter 6, this chapter addresses the following research questions:

- How do the depth of the available sleep mode and the target rate requirement determine the EE-optimal hardware configuration, and how do wake-up transition times and latency budgets constrain that choice?
- How can this coupling between sleep mode and transmit configuration be applied to schedule a BS serving bursty traffic with mixed QoS requirements?

7.3 Related Works

While this thesis pursues exact analytical optimization for single-link scenarios, recent literature has increasingly turned to data-driven and network-level approaches for managing sleep modes under uncertainty. Deep reinforcement learning assisted by digital twins has been proposed to anticipate performance degradation before deactivating BS components, mitigating the latency risk of deeper sleep states [78]. In multi-cell massive MIMO deployments, multi-agent reinforcement learning frameworks learn collaborative policies that jointly control antenna switching and multi-level sleep modes to minimize network-wide PC while managing inter-cell interference [79]. Beyond traditional cellular topologies, EE has been extended into cell-free Open-RAN architectures by moving from local BS control to end-to-end energy-aware orchestration of radio, optical fronthaul, and virtualized cloud processing resources [80].

Closer to the analytical perspective adopted here, prior work has minimized BS PC by either adjusting time-slot activation for single-antenna rush-to-sleep strategies [81] or numerically optimizing time, space, and power resources for single-band systems [82]. These approaches treat the active hardware configuration as essentially fixed and optimize when the BS should sleep. The contribution of this chapter is to instead let the active hardware configuration co-optimize with the sleep policy. The optimal (P, B, M) depends on the sleep depth available, and conversely the viable sleep depth depends on the latency budget and on how aggressively the BS can transmit during its active periods. This co-optimization is developed in Paper C for a single transmission instance, and is extended here to operation over a dynamic buffer.

7.4 Sleep-Mode Power Model and Wake-Up Latency

We adopt the sleep-mode framework of Section 5.2, briefly restated here for convenience. When the BS is not actively transmitting, it can power down parts of its hardware to conserve energy. The sleep PC is modelled as $P_s = \delta_s(\mu + D_0 M)$, where $\delta_s \in [0, 1]$ is the fraction of passive PC retained while sleeping.

Letting $\alpha \in [0, 1]$ denote the activity factor, defined as the proportion of time the BS is actively transmitting, the average PC follows (5.5),

$$\bar{P} = \alpha P_a + (1 - \alpha) P_s, \tag{7.1}$$

with P_a given by the multi-antenna PC model in (5.3). For a required average rate R, the activity factor satisfies

$$\alpha = \frac{R}{B \log_2(1 + \frac{MP\beta}{BN_0})}, \tag{7.2}$$

which couples the sleep policy directly to the active hardware configuration through the achievable transmission rate.

For the chapter-specific analysis we adopt the three sleep modes exemplified in 3GPP [83], ordered by increasing depth and increasing wake-up latency:

- **Idle mode**: $\delta_s = 1$, transition time $\tau_{\text{idle}} = 0$ ms. No power saved beyond active idle, no wake-up delay.
- **Deep sleep**: $\delta_s = \mu/(\mu + D_0 M)$, transition time $\tau_{\text{deep}} = 6$ ms. Per-antenna chains powered down, only the fixed circuitry μ remains.
- **Absolute sleep**: $\delta_s = 0$, transition time $\tau_{\text{abs}} = 50$ ms. All hardware powered down, but the largest wake-up time.

The transition time τ_k for sleep mode k is the wake-up latency the BS must absorb before it can resume active transmission, and it is the variable that couples the choice of sleep depth to the QoS constraints of the served traffic. Transitioning into a sleep state is assumed to be instantaneous [84], as only the wake-up direction carries a penalty.

7.5 Rush-to-Sleep and the Optimal Active Rate

When the required average rate R is lower than the optimal continuous rate R_{opt} obtained by Algorithm 1 in Chapter 6, the BS can achieve R by transmitting briefly at a high active rate and sleeping for the remainder of the time. This is the *rush-to-sleep* strategy, and its energy cost depends on which sleep mode is available.

For absolute and deep sleep, Paper C proves that the EE is maximized by always configuring the hardware to the unconstrained optimal configuration $(P_{\text{opt}}, B_{\text{opt}}, M_{\text{opt}})$ and scaling the activity factor linearly as $\alpha = R/R_{\text{opt}}$. The hardware configuration is decoupled from the rate requirement. This means that the BS transmits at the same rate it would have used under an infinite backlog, and simply sleeps for a longer fraction of the time when R is small. Average PC then grows linearly with R, and for absolute sleep the achieved EE matches the unconstrained EE upper bound for all $R \leq R_{\text{opt}}$.

A closed-form result holds specifically for deep sleep when the system is bandwidth-limited (i.e., the optimum lies on $B = B_{\max}$). In that case, the EE-optimal active configuration admits a solution in which the SNR is a numerical constant,

$$\mathrm{SNR}_{\mathrm{deep}} = u^* - 1, \qquad u^* = -\frac{2}{W_0(-2e^{-2})} \approx 4.92, \tag{7.3}$$

where $W_0(\cdot)$ is the principal branch of the Lambert W function. The corresponding active transmission rate

$$R_{\mathrm{deep}} = B_{\max} \log_2(u^*) \tag{7.4}$$

depends solely on the available bandwidth, and the optimal SNR of approximately 5.93 dB is the value at which the logarithm of the capacity expression begins to flatten relative to the linear PC growth in P and M. The corresponding spectral efficiency is $\log_2(u^*) \approx 2.30$ bit/s/Hz, which in 3GPP NR maps onto 16-QAM with a forward error correction coding rate of 0.58 [85].

This operating point is robust. In Chapter 6, the same constant emerged as the global optimum of the unconstrained EE problem in the regime where μ/B and $D_0 M/B$ are negligible. The deep-sleep result reproduces it exactly while retaining $D_0 > 0$ and $\mu > 0$, because μ is paid uniformly across active and sleep periods and therefore drops out of the objective. The convergence of the two derivations on the same SNR shows that 5.93 dB is independent of the channel gain β and of the target rate, and is in that sense a deployment-independent design target. An implication is that a macro cell with deep pathloss and a small cell with strong line-of-sight should both aim for the same operating SNR, dimensioning P and M via the optimal power-per-antenna ratio to reach it.

The intuition for why deep sleep and absolute sleep produce different active configurations is that, under absolute sleep, the fixed circuit power μ is paid exclusively during the active phase, which incentivizes the BS to transmit with higher power and more antennas to clear the buffer faster. Under deep sleep, μ is an unavoidable sunk cost across both states, and the BS relaxes into the 5.93 dB operating point. Consequently, $R_{\mathrm{abs}} > R_{\mathrm{deep}}$ and $\mathrm{SNR}_{\mathrm{abs}} > \mathrm{SNR}_{\mathrm{deep}}$ for any finite $B_{\max}$, although the two configurations converge to the same SNR as the bandwidth and power constraints are asymptotically relaxed.

For idle mode, the dependence of δ_s on M in the denominator of the EE objective prevents a closed-form solution, but the optimization can still be carried out by alternating the optimization of P and B with a line search over M, as described in Paper C. The optimal number of active antennas in idle mode grows with the rate requirement R, whereas it stays constant for absolute and deep sleep below their respective active rates.

7.6 Latency Constraints and Transition Penalties

The transmission deadline T is the latest moment by which the payload L must be delivered, a QoS constraint set by the application. The service interval $U \geq T$

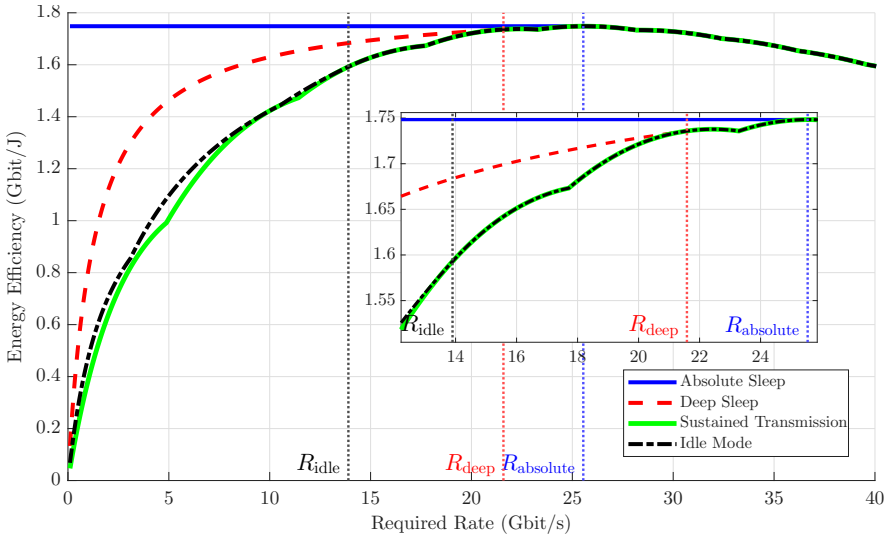


Figure 7.1: Energy efficiency versus required rate R for a BS capable of using absolute sleep, deep sleep, and sustained transmission. The activity factor α increases from 0 to 1 as R grows; the optimal point at $R = R_{\text{opt}}$ is highlighted.

is the duration before the BS must be active again for the next scheduled transmission. When $U = T$, the BS must be active again the moment the deadline expires; when $U > T$, the BS can finish transmitting early, enter sleep, and transition back to active within the remaining time. Wake-up from a sleep state takes time τ_k, so the available time for active transmission is bounded by both constraints.

We define the maximum permissible activity factor for sleep mode k as

$$\alpha_{\max,k} \triangleq \min\left(\frac{T}{U},\, 1 - \frac{\tau_k}{U}\right), \tag{7.5}$$

which reflects two competing physical constraints. The first is that the BS must finish transmitting before the deadline T, and the second is that it must reserve τ_k seconds to wake up before U. Sleep mode k is viable only if $\alpha_{\max,k} > 0$. The BS is constrained by whichever bottleneck occurs first, and it actively reaches this upper bound precisely when the latency constraints force it to transmit faster than its EE-optimal rate.

If the activity factor $\alpha = R/R_{\text{opt}}$ implied by the rush-to-sleep strategy is at most $\alpha_{\max,k}$, the transition delay is fully absorbed into the sleep period, and the analysis of the previous section carries over unchanged. If $\alpha > \alpha_{\max,k}$, however, the EE-optimal configuration is no longer feasible. The BS must instead use the entire available active window by setting $\alpha = \alpha_{\max,k}$, which forces the active rate

up to

$$R_{\text{high}} = \frac{R}{\alpha_{\max,k}}. \tag{7.6}$$

At this elevated rate the BS operates in a no-sleep mode governed by the closed-form expressions

$$P_{\text{high}} = \sqrt{c^{-1}\left(2^{R_{\text{high}}/B_{\max}} - 1\right)}, \qquad M_{\text{high}} = \frac{P_{\text{high}}}{\kappa(D_0 + \nu B_{\max})}, \tag{7.7}$$

with $c = \beta/(\kappa(D_0 + \nu B_{\max})B_{\max}N_0)$. The PC then grows exponentially in R_{high}, penalizing the total EC.

Paper C combines these ingredients into a single-instance selection procedure, Algorithm 2, which evaluates each of the three sleep modes for a given (L, T, U). A mode is discarded if $\alpha_{\max,k} \leq 0$. Otherwise the algorithm checks whether the rush-to-sleep activity factor fits within $\alpha_{\max,k}$, in which case the unconstrained optimal configuration applies, or whether the BS must transmit at R_{high} with the configuration in (7.7). The mode achieving the lowest EC is selected. Since the two cases exhaust the per-sleep-mode optimum and the final comparison spans all viable modes, the algorithm is globally EE-optimal for the single-instance problem. This single-instance procedure is where the scope of Paper C ends. It is, however, the building block on which the dynamic scheduler of Section 7.7 is constructed.

The interplay between L, T, and the available sleep depths is illustrated in Fig. 7.3, where $T = U$ is assumed, i.e., the next service interval begins the moment the transmission deadline expires (relaxing to $U > T$ favors absolute sleep across nearly all (L, T) combinations). Increasing T (more relaxed deadlines) and decreasing L (smaller payloads) shifts the optimum toward absolute sleep, while strict deadlines push the BS first into deep sleep, then into idle mode, and ultimately into the high-rate continuously-active regime where no sleeping occurs. The achieved EE is approximately constant along lines of constant rate $R = L/T$, indicating that the rate requirement is the primary determinant of both the achievable EE and the optimal sleep mode.

The role of the service interval is isolated in Fig. 7.2, which plots the EE achievable under each sleep mode as U sweeps from T to $2T$ for a fixed payload and deadline where idle mode is optimal at $U = T$. At $U = T$, idle mode wins because the deeper modes are forced to consume part of the deadline on their wake-up transitions, leaving them with shorter active windows and higher required transmit rates. As U grows, the additional slack amortizes the wake-up cost. Deep sleep overtakes idle mode shortly after $U = T$, and absolute sleep takes over the optimum near $U \approx 1.4T$ or around 95 ms. The mode selection in Fig. 7.3 is thus contested only in the regime where the service interval is tight relative to the transition times.

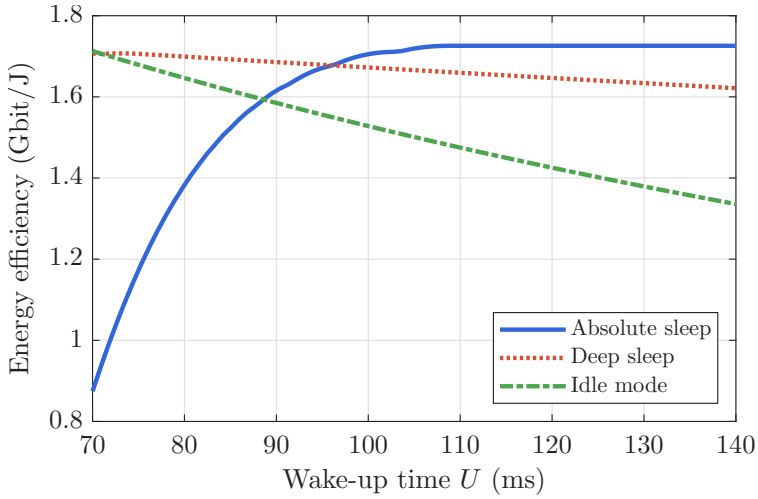


Figure 7.2: EE achievable under each sleep mode as a function of the service interval U.

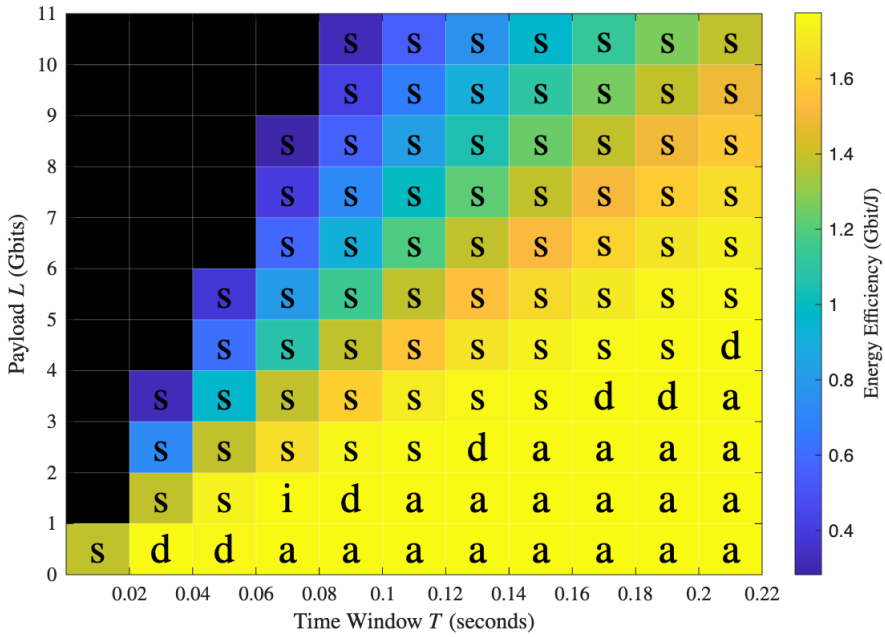


Figure 7.3: The EE and sleep mode selection for a BS for varying payload sizes and transmission deadlines. Black regions denote infeasible (L, T) combinations.

7.7 Dynamic Energy-Efficient Scheduling

Algorithm 2 configures a single transmission instance (L, T, U). A BS serving real traffic instead faces a buffer whose contents change as packets arrive and

deadlines pass. This section goes beyond the content in Paper C and applies the single-instance framework to that setting, treating it as a scheduling problem in which each decision epoch is reduced to an instance that Algorithm 2 can solve.

Each packet i is described by three instants and one size: it becomes available to the BS at its arrival time a_i, it carries a payload of L_i bits, and it must be fully delivered by its absolute deadline $T_i = a_i + \text{PDB}_i$, where PDB_i is the packet delay budget (PDB) of its QoS class. The interval $[a_i, T_i]$ is thus the window inside which packet i may be served, and the BS is free to finish its transmission anywhere within it. Let $\mathcal{N}(t)$ denote the set of packets that have arrived but not yet been delivered at time t, sorted so that $T_1 < T_2 < \cdots$. To avoid violating any deadline, the BS must sustain the cumulative rate

$$R_n(t) = \frac{1}{T_n - t} \sum_{i=1}^{n} L_i \tag{7.8}$$

for every n. The binding requirement is the bottleneck $n^\star = \arg\max_n R_n(t)$, and the subset $\{1, \dots, n^\star\}$ can therefore be treated as a single payload $L = \sum_{i=1}^{n^\star} L_i$ with deadline $T = T_{n^\star} - t$. This consolidation is what maps the buffer back onto the single-instance form of Section 7.6.

7.7.1 The Cost of Sleeping

The simplest policy is to wake as soon as the buffer is non-empty and clear the bottleneck subset immediately. Its weakness is visible in the gap cost. When the BS is inactive for a duration g, it can enter only the deepest mode whose transition time fits, so the energy spent is

$$C(g) = \begin{cases} (\mu + D_0 M)\, g, & g \le \tau_{\text{deep}}, \\ \mu\, g, & \tau_{\text{deep}} < g \le \tau_{\text{abs}}, \\ 0, & g > \tau_{\text{abs}}. \end{cases} \tag{7.9}$$

The same total inactive duration therefore costs nothing when collected into one interval longer than τ_{abs}, but costs the full passive PC when scattered into many intervals shorter than τ_{deep}. Clearing each packet on arrival leaves gaps of a few milliseconds between transmissions, none of them long enough to reach absolute sleep, so the hardware stays powered through intervals in which it does nothing useful. At low load this waiting dominates the EC.

A better approach is to let the buffer fill, where the deadlines allow it, and serve it in one run, so that the inactive time is collected in front of the burst rather than distributed through it. Deferring too far has the opposite energy cost. If the bottleneck rate (7.8) grows beyond R_{abs}, the BS is pushed into the high-rate configuration (7.7), whose PC grows exponentially in the required rate.

7.7.2 Choosing the Wake-Up Instant

Suppose the BS commits to a transmit rate R and expects to serve a set $\mathcal{B}$ of packets with cumulative payloads $\Lambda_n = \sum_{i=1}^{n} L_i$. Starting at time s is feasible only if every packet clears its deadline, that is, $\Lambda_n/R \leq T_n - s$ for all n, so the latest admissible instant is

$$s^\star(\mathcal{B}) = \min_{n \in \mathcal{B}} \left(T_n - \frac{\Lambda_n}{R} \right). \tag{7.10}$$

Equation (7.10) asks what start time a rate permits. Waking at $s^\star$ makes the preceding sleep interval as long as the deadlines allow, while guaranteeing that no rate above R is ever needed. The rate is fixed first at an EE-optimal value and the wake-up follows from it.

The wake-up instant is bounded from below by the arrivals themselves, since the BS cannot transmit a packet before it exists. The BS therefore wakes at $w = \max(a_1, s^\star(\mathcal{B}))$, where a_1 is the earliest arrival in the batch. When $s^\star > a_1$ the packets are deliberately left waiting in the buffer, which lengthens the preceding sleep interval. When $s^\star \leq a_1$ the deadlines are too tight to defer and the BS wakes on arrival. Note also that the later the BS wakes, the more packets will have arrived and joined the batch and thus $\mathcal{B}$ depends on w. The two are therefore evaluated alternately until they agree, which takes a few passes since deferring can only enlarge $\mathcal{B}$, and a larger $\mathcal{B}$ can only pull $s^\star$ earlier.

The rate itself comes from Algorithm 2. Once the wake-up instant is set and the batch consolidated by (7.8) into a payload L with deadline T, the service interval U is measured from the wake-up instant to the next planned wake-up, so that the sleep period that follows the burst is accounted for in the mode selection. The resulting instance (L, T, U) is passed to Algorithm 2, which returns the sleep mode and hardware configuration that are jointly optimal for it. The planner thus decides when to wake and what to group, while the theory of Paper C decides how to transmit. The complete policy is given in Algorithm 3. This is called lazy-wake batching because the BS will be lazy and try to sleep even after a packet has arrived to conserve energy and transmit packets in batches.

Here $L_{\mathcal{B}}$ denotes the cumulative payload of the current batch, so line 7 collects every packet arriving before the burst would finish. If packets arriving mid-burst raise the bottleneck rate above R, the BS falls back on the high-rate configuration (7.7) for the remainder of the run.

Fig. 7.4 shows an example of the resulting behavior for a mixed-traffic buffer containing four packets. A large packet with a short deadline is cleared at R_{high}, a subsequent smaller packet relaxes the requirement enough to transmit at R_{deep} and enter deep sleep, and later isolated arrivals leave enough room for absolute sleep. The BS moves between deep sleep and absolute sleep, generally avoiding waiting, sometimes utilizing the high-rate regime as the buffer evolves.

Algorithm 3 Lazy-Wake Batching

Require: Hardware and channel parameters $\beta, N_0, \mu, \nu, D_0, \kappa$ and limits $P_{\max}, B_{\max}, M_{\max}$ required by Algorithms 1 and 2; sleep modes $k \in \mathcal{K}$ with transition times τ_k and sleep powers $P_{s,k}$; packets of arrival time, size and deadline (a_i, L_i, T_i)

Ensure: Wake-up instant w, sleep mode k, and transmit configuration (P, B, M) for each cycle

compute the tangent rate R_k of every sleep mode k using Algorithm 1
$t \leftarrow 0$
while packets remain undelivered **do**
  $a_1 \leftarrow$ arrival time of the earliest undelivered packet
  $w \leftarrow a_1$, $R \leftarrow R_k$ of the deepest mode admitted by $w - t$
  repeat ▷ batch and wake-up instant agree
    $\mathcal{B} \leftarrow \{i : a_i \le w + L_{\mathcal{B}/R}\}$ ▷ arrivals during the burst
    $w \leftarrow \max(a_1,\ s^\star(\mathcal{B}))$ ▷ from (7.10)
  until w is unchanged
  consolidate $\mathcal{B}$ by (7.8) into a payload L with deadline T
  $U \leftarrow$ (next planned wake-up) $- w$
  $(k, P, B, M) \leftarrow$ Algorithm 2(L, T, U) ▷ coupled mode and configuration
  sleep in mode k on $[t, w)$
  transmit $\mathcal{B}$ in deadline order using (P, B, M); $t \leftarrow w + L/R$
end while

7.7.3 Bursty Traffic Evaluation

To evaluate the policy under realistic conditions, such as those mentioned in [8], we generate traffic with heavy-tailed payloads drawn from a bounded Pareto distribution, reproducing the empirical observation that a small fraction of sessions carries most of the data volume. Each packet is assigned a delay budget from the 5G QoS identifier (5QI) framework of 3GPP TS 23.501: roughly 300 ms for buffered video and background synchronization, 100 ms for streaming and web browsing, and 10 to 50 ms for ultra-reliable low-latency communication (URLLC) and real-time gaming. Arrivals follow an interrupted Poisson process (IPP) alternating between bursts and blanks of 0.5 to 8 seconds, and the average arrival rate λ is swept to vary the load.

Fig. 7.5 shows a Monte-Carlo simulation of the fraction of time the BS spends in each hardware state as the load grows. At low traffic intensities the long inter-arrival times let absolute sleep dominate the timeline. Increasing traffic rates make absolute sleep unavailable and shift the optimal pair toward deep sleep. Idle mode is rarely selected, since its only advantage over deep sleep is a zero transition time, and a policy built to manufacture long gaps removes the short-gaps mostly.

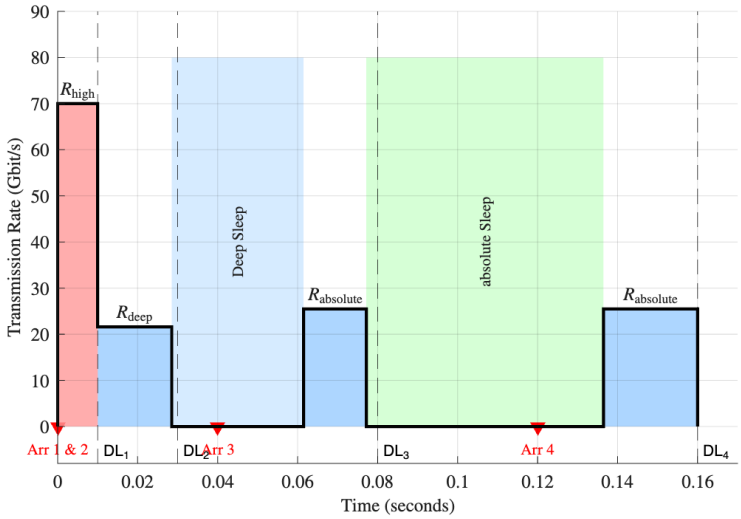


Figure 7.4: Sleep mode selection and active rate over time for a BS serving packets of mixed sizes and deadlines. Arrivals are shown in red and deadlines as dashed lines.

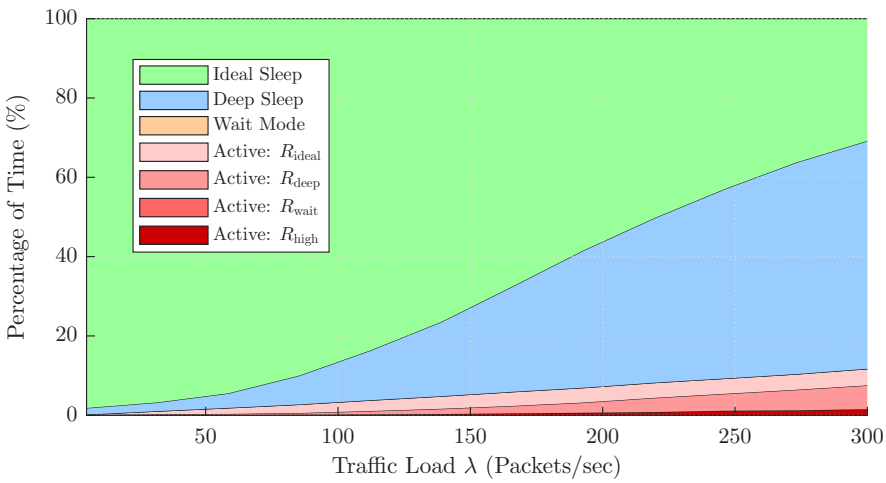


Figure 7.5: Percentage of time spent in each hardware state as the average traffic load λ grows.

Two things are worth stating. First, the wake-up instant in (7.10) is computed from arrivals that have not yet occurred, so Algorithm 3 is not causal. Therefore a deployed version would need to predict burst extent, for instance from the ON-state statistics of the arrival process or from machine learning. Second,

we do not claim that the resulting schedule is optimal, but it considers the important characteristics of the scheduling problem. Algorithm 3 is presented as evidence that the coupled sleep-mode and rate framework of Paper C extends to bursty traffic, and it can act as a baseline against which other schedulers can be compared.

7.8 Summary

The optimal active hardware configuration and the chosen sleep depth are coupled rather than independent. For absolute and deep sleep, the EE-optimal strategy at low rates is rush-to-sleep at the unconstrained optimal configuration $(P_{\text{opt}}, B_{\text{opt}}, M_{\text{opt}})$, with the activity factor scaling linearly with R. Deep sleep admits a closed-form active-rate optimum at the SNR $u^* - 1 \approx 5.93\,\text{dB}$, which reappears as the global optimum of the unconstrained problem and is therefore independent of β and of the target rate. Strict deadlines and finite wake-up times bound the viable activity factor through $\alpha_{\max,k}$, and when this bound binds the BS is forced into the high-rate regime where PC grows exponentially in the required rate.

Applying this to bursty traffic, we showed that the efficiency of the scheduling lies not only in its transmission configuration but also in how its inactive time is distributed. Because the gap cost (7.9) vanishes beyond τ_{abs}, inactive seconds scattered between packets are expensive. Deferring the wake-up to the latest feasible instant (7.10) and letting Algorithm 2 select the coupled sleep mode and rate for the resulting batch reduces EC while meeting every deadline. The framework of Paper C therefore extends from a single transmission to a dynamic buffer.

8 Energy-Efficient Dual-Band Communication: Managing Sub-THz Carriers and Sleep Modes

8.1 Introduction

Chapters 6 and 7 characterized the EE of a single-band BS, treating the joint configuration of transmit power, bandwidth, and antennas, and then layering advanced sleep modes and latency-constrained scheduling on top. In all of that analysis, the BS has access to one carrier with a fixed bandwidth B. However, as wireless data traffic grows toward the rate targets envisioned for 6G, networks are transitioning toward higher frequencies; even as high as sub-Terahertz (sub-THz) to access the bandwidth that sub-6 GHz cannot provide [86, 87]. The RAN already accounts for the vast majority of the electricity consumed by mobile operators [21], and adding a second high-bandwidth carrier on top of the existing infrastructure will increase this even further.

A single bandwidth-limited carrier cannot absorb arbitrary rate growth without paying an exponential penalty in transmit power, as the analysis made explicit in the previous chapter. Adding a second carrier with much larger bandwidth removes that exponential wall at the cost of a static power penalty μ_2 that must be paid whenever the secondary band is even powered on. Whether the secondary band is worth activating therefore depends on where the BS sits along the rate axis and on how much of the time it would actually need to use the additional bandwidth. While dual-band carrier aggregation is conventionally motivated by peak throughput [87], this chapter studies it instead as a PC minimization problem and derives the rate thresholds at which activating the secondary carrier becomes EE-optimal.

8.2 Research Questions

This chapter addresses the following research questions:

- At what required data rate does it become EE-optimal to awaken and utilize a secondary sub-THz carrier in a dual-band system?
- When both bands are active, how should traffic be allocated and how should the BS hardware be configured in each band to minimize total PC?

8.3 Related Work

Single-band sleep-mode strategies and analytical EE limits have been studied in various forms, including time-domain rush-to-sleep policies [81] and joint numerical optimization across time, space, and power [82]. Foundational PC models for aggregated carriers were introduced in [65], who provide measurement-aligned per-band constants across sub-6 GHz and higher-band massive MIMO transceivers; we adopt their per-carrier decoupling as the basis for the multi-carrier PC model of Section 5.1.4.

Where this work departs from the prior literature is in treating sleep-mode coordination across two non-contiguous bands as the central object of analysis. Carrier aggregation has been extensively studied for maximizing peak throughput, but the question of when a BS should keep a high-bandwidth secondary carrier asleep, and at what rate it should awaken it, has not been answered analytically. The contribution of Paper D, summarized in this chapter, is precisely to derive the closed-form rate thresholds and asymptotic allocation rules that govern this decision.

8.4 Contributions

By jointly optimizing the transmit power, number of active antennas, and sleep-mode activity factors of both bands, Paper D identifies four distinct operational regions that govern the EE-optimal behavior across the entire rate spectrum. The boundaries between these regions are characterized in closed form via the Lambert W function: a saturation rate R_1^* at which the primary band first reaches continuous operation, an activation threshold R_{bp} at which the secondary band should awaken from sleep, a second saturation rate R_{sat} at which the secondary band itself reaches continuous operation, and a crossover rate R_{cross} above which the dual-band architecture strictly outperforms the single-band baseline in EE. In the asymptotic high-rate limit, the optimal traffic split between the two bands becomes independent of the channel conditions and is dictated entirely by the ratio of their bandwidths and PA efficiencies.

8.5 System Model

We consider a dual-band BS communicating with a UE over two widely-spaced carriers indexed by $i \in \{1, 2\}$. Band 1 is a coverage-oriented sub-6 GHz primary carrier with relatively favorable propagation, high PA efficiency, and narrow bandwidth B_1. Band 2 is a capacity-oriented sub-THz secondary carrier with severe per-antenna pathloss, lower PA efficiency, and large bandwidth $B_2 \gg B_1$. The system model is illustrated in Fig. 8.1.

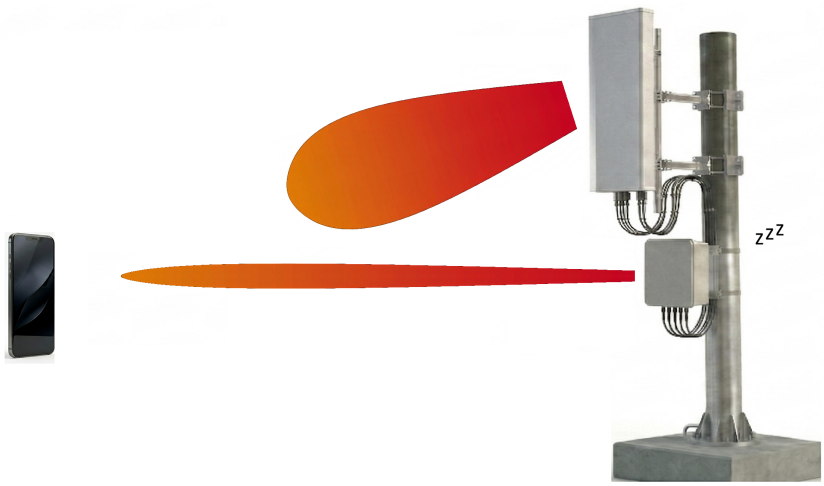


Figure 8.1: System model of the dual-band BS downlink. The multi-antenna BS coordinates transmissions to a UE across two frequency bands, utilizing sleep modes and waking the secondary band only as needed.

For each band, the BS uses M_i transmit antennas and the UE uses N_i receive antennas. Under far-field LoS propagation with perfect CSI, maximum ratio transmission (MRT) at the BS combined with maximum ratio combining (MRC) at the UE yields a total array gain of $M_i N_i$, so the active-mode rate in band i is

$$R_i = B_i \log_2\left(1 + \frac{P_i M_i N_i \beta_i}{B_i N_0}\right). \tag{8.1}$$

The site-level PC follows the multi-carrier extension of Section 5.1.4 with $K = 2$, combined with the per-band sleep-mode framework of Section 5.2. Each band carries its own active PC

$$P_{a,i} = \frac{P_i}{\kappa_i} + \mu_i + (D_{0,i} + \nu_i B_i) M_i, \tag{8.2}$$

its own sleep PC $P_{s,i} = \mu_i$ (only the band-specific fixed circuitry remains), and

its own activity factor $\alpha_i \in [0, 1]$. The total site PC is

$$\bar{P}_{\text{site}} = \sum_{i=1}^{2} \left[\alpha_i P_{a,i} + (1 - \alpha_i) P_{s,i}\right]. \tag{8.3}$$

Per-band activity factors are what make the activation threshold derived below a meaningful quantity: awakening Band 2 carries an isolable static cost μ_2 that the primary band cannot share, because the two carriers operate at widely separated frequencies and use distinct hardware components.

The objective is to deliver a required total rate $R = R_1 + R_2$ while minimizing $\bar{P}_{\text{site}}$.

8.6 Single-Band Recap and Region 1

Before treating the dual-band problem, we recall the single-band optimum from Chapter 6. For each band considered in isolation, the EE-optimal hardware configuration satisfies the power-per-antenna ratio

$$M_i^* = \frac{P_i^*}{\kappa_i (D_{0,i} + \nu_i B_i)}, \tag{8.4}$$

with optimal transmit power

$$P_i^* = \sqrt{(u^* - 1)/c_i}, \qquad c_i = \frac{N_i \beta_i}{\kappa_i (D_{0,i} + \nu_i B_i) B_i N_0}, \tag{8.5}$$

where $u^* = -2/W_0(-2e^{-2}) \approx 4.92$ is the same numerical constant that governed the deep-sleep result in Chapter 7. The corresponding active rate is $R_i^* = B_i \log_2(u^*)$ and the active-mode EE is $\text{EE}_{i,\text{opt}} = R_i^*/(2P_i^*/\kappa_i + \mu_i)$.

Because Band 1 enjoys more favorable propagation and higher hardware efficiency, $\text{EE}_{1,\text{opt}} > \text{EE}_{2,\text{opt}}$ holds throughout this chapter. As a result, for any low rate the BS is best served by the primary band alone:

Region 1 (Primary-Band Duty-Cycling, PBDC). For $0 < R \le R_1^*$, Band 2 stays asleep ($\alpha_2 = 0$), Band 1 transmits at (P_1^*, M_1^*), and the activity factor $\alpha_1 = R/R_1^*$ scales linearly with the rate requirement. Average PC therefore grows linearly in R.

8.7 The Saturation Region and the Activation Threshold

Once R exceeds R_1^*, Band 1 cannot duty-cycle further: it must either be operated continuously at higher power than P_1^*, or the BS must awaken Band 2 and pay the static penalty μ_2. Whether the second option is preferable depends on the marginal cost of pushing one extra bit through each band.

Region 2 (Primary-Band Saturation, PBS). For rates just above R_1^*, Band 1 is fully active ($\alpha_1 = 1$) and the transmit power P_1 must rise above P_1^* to deliver the required rate. Solving the rate equation for P_1 yields

$$P_1(R_1) = \sqrt{c_1^{-1}(2^{R_1/B_1} - 1)}, \tag{8.6}$$

so the average PC of Band 1 grows exponentially in the rate requirement. Band 2 remains asleep because the marginal cost of additional Band 1 power is still smaller than the static cost μ_2 of waking the secondary band.

The boundary between Region 2 and Region 3 is the rate at which these marginal costs equalize. This is the central analytical result of the chapter:

Theorem 2 (Secondary-band activation)**.** *Suppose* $\mathrm{EE}_{1,\mathrm{opt}} > \mathrm{EE}_{2,\mathrm{opt}}$. *The dual-band EE is maximized by activating Band 2 (i.e.,* $\alpha_2 > 0$*) if and only if the required rate* R *exceeds the threshold*

$$R_{\mathrm{bp}} = B_1 \log_2(1 + c_1 P_{1,\mathrm{bp}}^2), \tag{8.7}$$

where the activation power in Band 1 is the closed-form quantity

$$P_{1,\mathrm{bp}} = \xi + \sqrt{\xi^2 - 1/c_1}, \qquad \xi = \frac{\kappa_1 B_1 \sqrt{u^* - 1}}{\kappa_2 B_2 \sqrt{c_2}\, \ln(u^*)}. \tag{8.8}$$

At the threshold, $\alpha_1 = 1$.

The proof, given in full in Paper D, follows from equating the marginal PC derivatives of the two bands. In Band 1, increasing R_1 requires increasing P_1 along (8.6), giving $dP_{\mathrm{avg},1}/dR_1 = (1 + c_1 P_1^2) \ln 2/(\kappa_1 B_1 c_1 P_1)$. In Band 2, increasing R_2 is achieved by raising the activity factor α_2 at the EE-optimal active power P_2^*, giving $dP_{\mathrm{avg},2}/dR_2 = 2\sqrt{u^* - 1}/(\kappa_2 B_2 \sqrt{c_2} \log_2(u^*))$. Setting these equal produces a quadratic in P_1, whose positive root is (8.8).

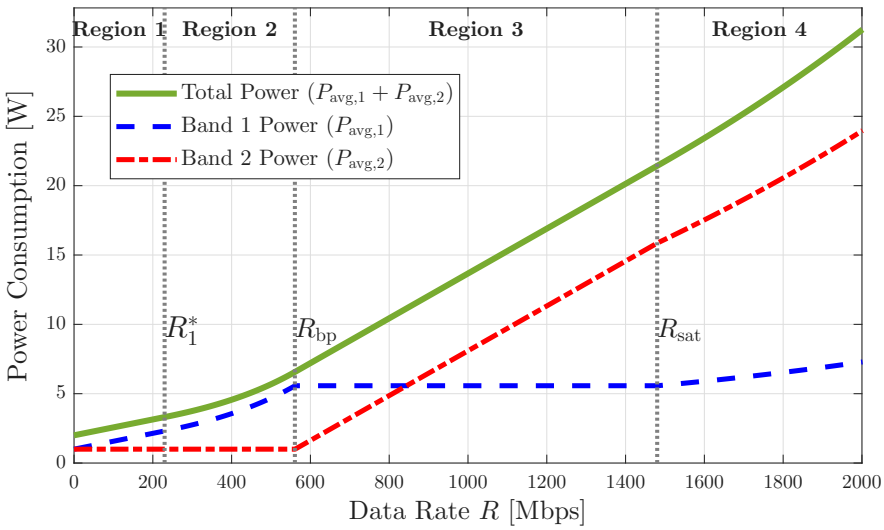
Region 3 (Secondary-Band Duty-Cycling, SBDC). For $R_{\mathrm{bp}} < R \le R_{\mathrm{sat}}$, Band 1 is fully active at the fixed power $P_{1,\mathrm{bp}}$ and contributes the constant rate R_{bp}. The residual rate $R - R_{\mathrm{bp}}$ is offloaded to Band 2, which transmits at its EE-optimal power P_2^* with activity factor $\alpha_2(R) = (R - R_{\mathrm{bp}})/(B_2 \log_2(u^*))$. Total PC returns to a linear scaling in R once Band 2 takes over the residual traffic.

Region 4 (Dual-Band Saturation, DBS). The saturation rate $R_{\mathrm{sat}} = R_{\mathrm{bp}} + B_2 \log_2(u^*)$ is the point at which Band 2 itself reaches $\alpha_2 = 1$. Beyond it, both bands are continuously active and must increase their transmit powers above the EE-optimal values, so PC grows exponentially again.

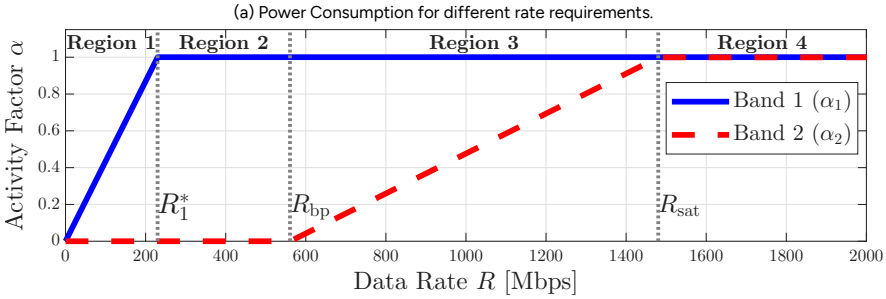
The four-region structure is summarized in Table 8.1, and the resulting PC and activity-factor curves are shown in Fig. 8.2. The curve demonstrates a linear scaling in Region 1, an exponential scaling in Region 2, a return to linear scaling in Region 3 as Band 2 absorbs the residual rate, and exponential scaling again in Region 4.

Table 8.1: Dual-Band Operation Regions for Band 1 and Band 2 Carriers.

Region	Rate R	Band 1 Status	Band 2 Status
1 (PBDC)	$0 < R \leq R_1^*$	active ($\alpha_1 < 1$)	Off ($\alpha_2 = 0$)
2 (PBS)	$R_1^* < R \leq R_{\text{bp}}$	active ($\alpha_1 = 1$)	Off ($\alpha_2 = 0$)
3 (SBDC)	$R_{\text{bp}} < R \leq R_{\text{sat}}$	active ($\alpha_1 = 1$)	active ($\alpha_2 < 1$)
4 (DBS)	$R > R_{\text{sat}}$	active ($\alpha_1 = 1$)	active ($\alpha_2 = 1$)



(a) Power Consumption for different rate requirements.



(b) Optimal activity factors α_1 and α_2 for different rate requirements.

Figure 8.2: Dual-band BS power consumption (a) and activity factors (b) across varying data rate requirements. The four operational regions PBDC, PBS, SBDC, and DBS are visible from left to right.

8.8 Asymptotic Traffic Allocation in Region 4

In the dual-band saturation regime, both activity factors are pinned at one and only the per-band transmit powers (P_1, P_2) remain as optimization variables.

Forming the Lagrangian of the resulting problem and writing the KKT conditions gives

$$\frac{1+c_1P_1^2}{\kappa_1 B_1 c_1 P_1} = \frac{1+c_2P_2^2}{\kappa_2 B_2 c_2 P_2}, \tag{8.9}$$

which expresses that the marginal PC must be equal across the two bands at the optimum. No closed-form solution exists for finite R, but the asymptotic limit admits a clean structural result.

Theorem 3 (Asymptotic traffic split)**.** *In Region 4, as $R \to \infty$,*

$$\lim_{R\to\infty} \frac{P_1(R)}{P_2(R)} = \frac{\kappa_1 B_1}{\kappa_2 B_2}. \tag{8.10}$$

The proof rearranges (8.9) as $P_1/P_2 = \kappa_1 B_1/(\kappa_2 B_2) + O(1/P_2^2) - O(1/(P_1P_2))$ and observes that the residual terms vanish as $P_1, P_2 \to \infty$. The interpretation is that, in the very high-rate regime, the optimal traffic split is dictated entirely by the the PA efficiencies and the available bandwidths.

8.9 Crossover with the Single-Band Baseline

Theorem 2 identifies when activating Band 2 is optimal *within* the dual-band system, but it does not by itself establish when the dual-band architecture outperforms a single-band baseline. The latter question depends on the static cost μ_2 of running the secondary circuitry at all. We define the crossover rate R_{cross} as the rate above which the dual-band system has lower average PC than a Band-1-only baseline that pushes through the saturation regime alone.

Theorem 4 (Dual-band crossover)**.** *The crossover rate is the solution of*

$$\frac{2}{\kappa_1\sqrt{c_1}}\sqrt{2^{R/B_1}-1} = \frac{2P_{1,\text{bp}}}{\kappa_1} + \frac{R-R_{\text{bp}}}{R_2^*}\left(\frac{2P_2^*}{\kappa_2}\right) + \mu_2, \tag{8.11}$$

which admits the approximate closed-form solution

$$R_{\text{cross}} \approx -C\left(\frac{\kappa_2 R_2^*}{2P_2^*}\right) - \frac{2B_1}{\ln 2}\, W_{-1}(Z), \tag{8.12}$$

where $W_{-1}(\cdot)$ is the lower branch of the Lambert W function, C collects the static and offset terms of (8.11)*, and Z is the corresponding argument given in Paper D.*

The proof equates the saturated single-band PC on the left side of (8.11) to the dual-band PC in Region 3 on the right side, applies the high-rate approximation $2^{R/B_1} - 1 \approx 2^{R/B_1}$, and rewrites the equation in the canonical form $Ye^Y = Z$ that the Lambert W function inverts. The lower branch W_{-1} is the relevant branch because the argument is strictly negative.

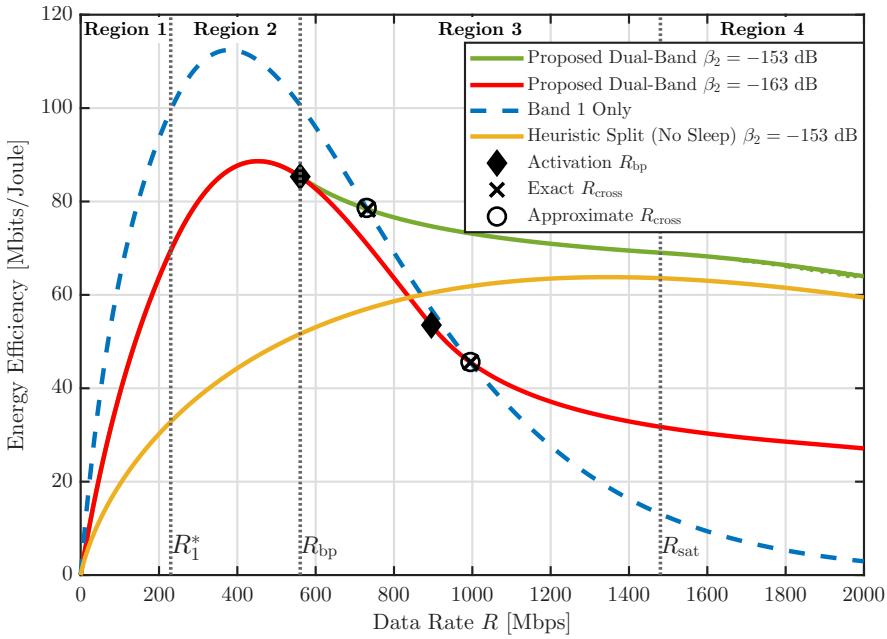


Figure 8.3: EE of the proposed dual-band BS for different rate requirements, compared against a Band-1-only baseline and a heuristic bandwidth-proportional split. The crossover rate R_{cross} from Theorem 4 marks where the dual-band scheme strictly outperforms the baseline. A 10 dB Band 2 blockage shifts R_{cross} to higher rates.

The qualitative content of Theorem 4 is that activating Band 2 is EE-optimal only once the exponential PC growth on Band 1 exceeds the static penalty μ_2. The rate at which this happens is sensitive to the propagation conditions in Band 2: if Band 2 suffers an additional 10 dB of pathloss due to blockage or imperfect CSI, both R_{bp} and R_{cross} shift to higher rates and the BS is forced to rely on Band 1 longer. Fig. 8.3 shows the EE comparison between the proposed dual-band scheme, a Band-1-only baseline, and a heuristic bandwidth-proportional split, with and without Band 2 blockage. The exact and approximate values of R_{cross} from Theorem 4 coincide very well with the numerical crossing point.

8.10 When the Sub-THz Baseline Eclipses the Sub-6 GHz Baseline

A related question, not treated in Paper D itself but answerable with the same machinery, is at what rate a Band-2-only system would outperform a Band-1-only system if both were used in isolation. This is structurally similar to Theorem 4 but compares the two single-band baselines rather than a single-band baseline against the dual-band scheme.

Theorem 5 (Single-band baseline crossover)**.** *Let the single-band Band 1 and*

Band 2 baselines each operate with optimal sleep modes. The rate at which the Band 2 baseline becomes strictly more energy-efficient than the Band 1 baseline is approximately

$$R_{\text{base}} \approx -\tilde{C}\left(\frac{\kappa_2 R_2^*}{2P_2^*}\right) - \frac{2B_1}{\ln 2}\, W_{-1}(\tilde{Z}), \tag{8.13}$$

where $\tilde{C} = \mu_2 - \mu_1$ *and* $\tilde{Z} = -\frac{\kappa_2 R_2^* \ln 2}{2P_2^* B_1 \kappa_1 \sqrt{c_1}} \exp\left(-\frac{\tilde{C}\kappa_2 R_2^* \ln 2}{4P_2^* B_1}\right)$.

The proof follows the same template as Theorem 4: at the crossing point, Band 1 has been forced into its saturated exponential regime while Band 2 still uses sleep modes by scaling $\alpha_2 < 1$, so equating the two PC expressions and applying the Lambert W inversion gives the result. The structural difference between R_{base} and R_{cross} is that the former compares two single-band systems each paying a single fixed cost (μ_1 versus μ_2), while the latter compares a single-band system against a dual-band system that pays both fixed costs simultaneously plus the offset $2P_{1,\text{bp}}/\kappa_1$ from running Band 1 at its activation power. Consequently $R_{\text{cross}} > R_{\text{base}}$ in general because it takes more rate to justify turning a second band on rather than allocating traffic to an already existing band.

8.11 Summary

This chapter established the EE limits of a dual-band BS combining a coverage-oriented sub-6 GHz primary carrier with a high-bandwidth sub-THz secondary carrier. The joint optimization of (P_i, M_i, α_i) across both bands produces a four-region operational structure: linear PC growth while the primary band duty-cycles, exponential growth as it saturates, linear growth again once the secondary band absorbs the residual rate, and exponential growth in the final dual-saturation regime. The boundaries between these regions are characterized in closed form. The activation threshold R_{bp} in Theorem 2 marks the rate at which the marginal PC of pushing harder on Band 1 first equals the marginal PC of waking Band 2. The crossover rate R_{cross} in Theorem 4 marks the rate above which the entire dual-band architecture outperforms the single-band baseline, and is sensitive to Band 2 blockage. In the very high-rate asymptotic regime, Theorem 3 shows that the optimal traffic split between the bands becomes independent of channel conditions and is dictated entirely by the ratio of bandwidths and PA efficiencies. Together, these results provide closed-form design rules for sleep-mode coordination across dual-band deployments.

9 Conclusions and Future Works

9.1 Conclusions

The rapid growth in data rates poses challenges for a network evolution in which EC scales sustainably in the future. In this thesis, we have explored the fundamentals of EE optimization for wireless communication links from both the UE and the BS perspective, addressing the energy cost of configuring an RIS, the control signaling required to operate it, the joint hardware optimization of multi-antenna BSs, and the coordination of advanced sleep modes across single- and dual-band sites. We now revisit the four research questions posed in Section 1.4 and state the answers that this thesis provides to each of them.

RQ1: *Can the total EC of a UE be reduced by using an RIS, if we account for the EC required to estimate the channel and configure it? How can the EC of the UE be reduced by optimizing pilot sequence length and transmit power while configuring the RIS according to the payload length?*

The answer to the first part is yes, but only if the configuration granularity is adapted to the payload. Paper A established that, for an RIS-aided UE transmitting short payloads, the pilot overhead required to configure the surface is a limiting factor in minimizing EC. By grouping RIS elements into subarrays with a common phase shift and jointly tuning the subarray size and pilot power, the EC can be significantly reduced compared to the baseline where each element is configured individually. The optimal configuration depends on the payload length and the relative strength of the direct and RIS-assisted paths: short payloads and strong direct paths (i.e., a weak direct path loss compared to the path loss via the RIS) favor large subarrays with low pilot power, while long payloads and weaker direct paths favor finer per-element control with higher pilot power. Imperfect CSI due to low pilot power leads to errors in the phase-shift config-

urations, which in turn requires increased data-signal power to reach the SNR target. The percentage of total energy allocated to pilot signaling at the optimum lies in the range of $10–40\%$.

RQ2: *How can the wireless control signaling overhead required to feed back phase-shift configurations from the BS to the RIS be minimized while maintaining the SNR?*

The overhead can be reduced from linear to logarithmic scaling in the number of RIS elements at a bounded SNR cost. Paper B addressed the wireless control signaling required to feed back the computed RIS configuration to the surface. By exploiting the geometric structure of the cascaded LoS channel, we designed a quantization codebook for which the feedback overhead scales only logarithmically with the number of RIS elements, while guaranteeing that the reflected beam remains within the half-power beamwidth. The design extends naturally to three-dimensional planar RIS geometries and to channels with a static path, and we showed that under user mobility a differential feedback scheme of only a few bits per update is sufficient to track the channel, with the static-path alignment phase being the term that requires the most frequent updates. The combined effect of finite feedback resolution and discrete phase-shift hardware can be characterized analytically and managed by matching the static-path alignment resolution to the bits per RIS element hardware phase resolution.

RQ3: *For a BS, what is the optimal joint configuration of transmit power, bandwidth, and number of antennas to maximize EE? Furthermore, how do the sleep mode-specific transition times dictate this optimal hardware configuration?*

The optimal configuration is characterized by a set of ratios that hold at the optimum, but the available sleep depth shifts which operating point the BS should aim for. Paper C analyzed the EE of multi-antenna BSs using a comprehensive transceiver PC model. We showed that in the absence of constraints on power or bandwidth, the power-to-bandwidth ratio converges to an optimal value, so that an excess of bandwidth does not by itself lead to improved EE. When the system is bandwidth-limited, there exists an optimal power-per-antenna ratio dictated solely by the hardware-characterizing constants, with the radiated power equal to the passive transceiver PC at the optimum. Together these two ratios fix the operating SNR to the numerical constant $u^* - 1 \approx 5.93\,\text{dB}$, corresponding to a spectral efficiency of roughly $2.3\,\text{bit/s/Hz}$. This value is independent of the pathloss and of the hardware constants, and the same constant reappears as the global optimum of the unconstrained problem, which makes it a deployment-independent target that a BS should be dimensioned to reach. We designed an alternating-optimization algorithm that rapidly converges to the joint EE optimum over transmit power, bandwidth, and number of antennas.

The framework was then extended to incorporate advanced sleep modes and

packet-level latency constraints. When the required average rate falls below the EE-optimal rate, the BS should apply a rush-to-sleep strategy and transmit at the EE-optimal rate for a fraction of the time and sleeps for the remainder, so that the average PC grows linearly with the required rate while the achieved EE stays at its upper bound. The optimal transmit rate depends on the sleep depth available. Under absolute sleep depth power is consumed only while active, which dictates that the BS should transmit at a higher rate and with more antennas, whereas under deep sleep, where passive circuit power is always consumed, the BS should use the $5.93\,\mathrm{dB}$ SNR operating point. Sleep depth and transmit rate are therefore a coupled optimization pair rather than two independent choices. We further characterized how strict deadlines and finite wake-up transition times limit how deep a sleep state the BS can afford to enter, and proposed a dynamic event-driven scheduling algorithm that continuously balances the energy savings of deeper sleep against the energy penalty of compensating for wake-up delays. Stochastic traffic simulations confirmed that under realistic bursty arrival patterns, the deeper sleep modes are used the most under low traffic intensity periods.

RQ4: *For a dual-band BS combining a coverage-oriented sub-6 GHz carrier with a capacity-oriented high band carrier, what operational regimes govern their joint optimal behavior? How should traffic be allocated between the two bands?*

The joint optimum is determined by the required rate and dictated by four operating regions, traversed as the demand grows, with the transitions between them given in closed form. At low rates only the primary band is used and it duty-cycles between its EE-optimal configuration and sleep. Therefore the average PC grows linearly with the rate. Once that band is continuously active, further demand can only be met by raising its transmit power, which is expensive because the band is bandwidth-limited. This cost eventually justifies turning on the secondary band. Activation is EE-optimal precisely when the cost of pushing more traffic through the saturated primary band exceeds the static cost of turning the secondary band on. We derived this activation threshold in closed form via the the Lambert W function. Beyond it, the additional traffic is offloaded to the secondary band, which duty-cycles in turn until it too saturates. In the high-rate limit the optimal split is dictated by the ratio of the two bandwidths and PA efficiencies in the two bands.

Together, these results point to a unifying principle. Namely that the active hardware and resources should be matched to the actual traffic demand. Whether that means deciding how much pilot energy to spend on configuring an RIS for a short payload, how many bits to that are needed to feed back the resulting configuration, how much power or antennas to use, how to enter a sleep state between data packet transmissions, or whether to wake a secondary carrier at all. The thesis develops this principle into closed-form design rules at each of these scales.

9.2 Future Works

The analysis carried out in this thesis suggests several natural directions for further investigation.

- **Extension of the fundamental scaling behaviors to richer channel and system models.** The closed-form characterizations of optimal power-to-bandwidth and power-to-antenna ratios were derived under far-field LoS-dominant assumptions. Extending the analysis to wideband multi-path channels, OFDM-based waveforms, and multi-user scenarios with spatial correlation would clarify how the structural rules carry over to deployments closer to current 5G systems. A multi-receive-antenna extension is a particularly natural next step, as is the joint treatment of more than two aggregated carriers.
- **RIS-aided systems beyond the single-link, single-RIS setting.** The subarray-based pilot design and the logarithmic-feedback codebook were both developed for a single UE served by a single RIS. Extending these techniques to multi-user, multi-RIS, and multi-antenna BS/UE deployments raises new questions about how subarray sizes and codebook indices could be coordinated across surfaces, and how the resulting control signaling should share the radio control channel with other users. Joint optimization of EC at the UE, BS, and RIS is an open problem.
- **Mobility, blockage, and channel aging in feedback design.** The differential feedback scheme was analyzed under a smooth mobility model. More realistic settings, including sudden blockage events, multi-cluster propagation, and finite RIS reconfiguration latency, would test how robust the few-bits-per-update design is and how often the codebook itself should be refreshed. Learning-based approaches that adapt the codebook to the local propagation environment are a promising direction.
- **Sleep-mode management under realistic traffic and learning-based scheduling.** Joint user scheduling and sleep-mode scheduling at the cell level is a natural extension. Validating and refining the scheduler against measurement-driven traffic traces, and incorporating learned arrival predictors that anticipate future bursts would increase EE.
- **Dual-band and multi-band coordination beyond the two-carrier case.** The four operational regions identified in Paper D arise from a primary/secondary carrier pair. A site combining even more carriers — for example, a low band for coverage, a mid band for capacity, and a sub-THz band for peak throughput — would generate a richer region structure and more activation thresholds. The integration of RIS into the dual-band analysis when the secondary carrier suffers from blockage could also be analyzed.

- **Validation against measurement and prototype hardware.** The analytical results in this thesis depend on hardware-characterizing constants that are not publicly disclosed and that vary across vendors and product generations. Calibrating the models to specific BS and RIS prototypes, and comparing the predicted EE-optimal operating points against measured power consumption, would both validate the analytical framework and quantify the gap between fundamental limits and current implementations.

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Appended papers